

CS 305: Computer Networks

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Lecture 6: Transport Layer

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Chapter 2: outline

2.1 principles of network applications

2.2 Web and HTTP

2.3 electronic mail

- SMTP, POP3, IMAP

2.4 DNS

2.5 P2P applications

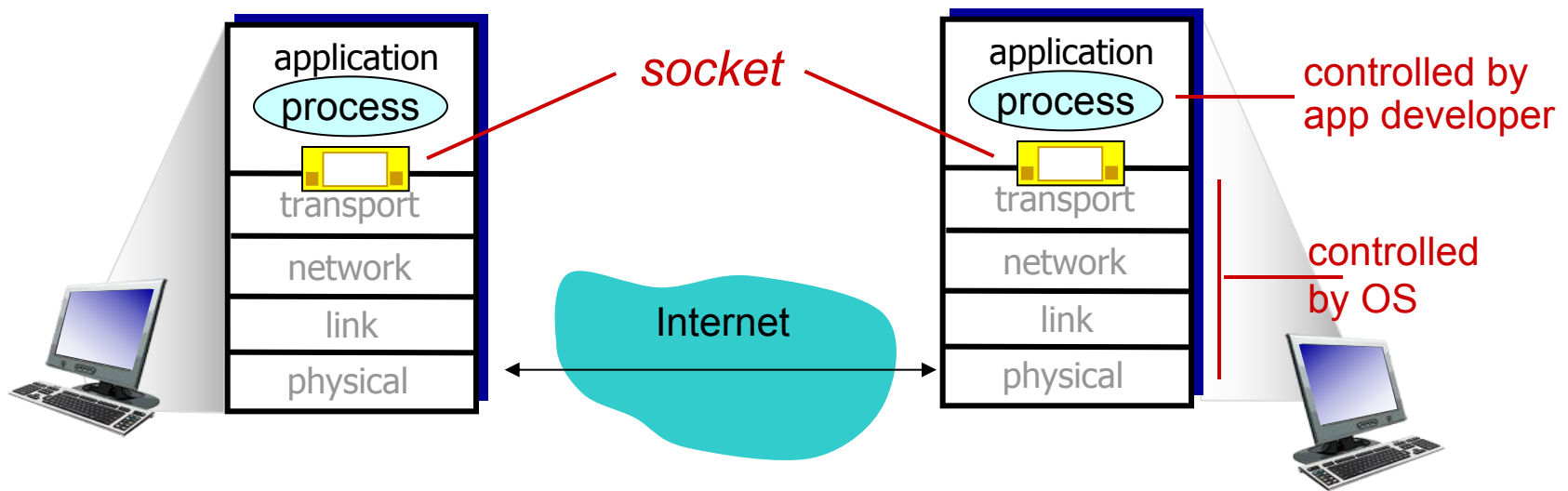
2.6 video streaming and content distribution networks

2.7 socket programming with UDP and TCP

Socket programming

Goal: learn how to build client/server applications that communicate using sockets

Socket: door between application process and end-end-transport protocol



Socket programming

Socket programming: how we can use socket API for creating communication between client and server processes.

Two socket types for two transport services:

- **UDP:** unreliable datagram
- **TCP:** reliable, connection-oriented

Application Example:

1. client reads a line of characters (data) from its keyboard and sends data to server
2. server receives the data and converts characters to uppercase
3. server sends modified data to client
4. client receives modified data and displays line on its screen

Socket programming with UDP

UDP: no “connection” between client & server

- ❖ no handshaking before sending data
- ❖ sender explicitly attaches destination IP address and port number to each packet
- ❖ receiver extracts sender IP address and port number from received packet

UDP: transmitted data may be lost or received out-of-order

Application viewpoint:

- ❖ UDP provides *unreliable* transfer of groups of bytes (“datagrams”) between client and server

Client/server socket interaction: UDP

server (running on *serverIP*)

create socket, port= x:
serverSocket =
socket(AF_INET,SOCK_DGRAM)

read UDP datagram from
serverSocket

write reply to *serverSocket*
specifying client address,
port number

client

create socket:
clientSocket =
socket(AF_INET,SOCK_DGRAM)

Create datagram with server IP and
port=x; send datagram via
clientSocket

read datagram from
clientSocket

close
clientSocket

Segment: the transport-layer packet for TCP
Datagram: the packet for UDP

Example app: UDP server

Python UDPServer

include Python's socket library → `from socket import *`

`serverPort = 12000`

create UDP socket → `serverSocket = socket(AF_INET, SOCK_DGRAM)`
" 表示绑定所有本地 IP 地址。"
IPv4
UDP socket
UDP socket is identified by destination IP address and port number

bind socket to local port number 12000 → `serverSocket.bind(('', serverPort))`
指定服务器的 IP 地址和端口号。

`print ( "The server is ready to receive")`

loop forever → `while True:`
每次发送时都需要告诉系统要发给谁。

Read from UDP socket into message, getting client's address (client IP and port) → `message, clientAddress = serverSocket.recvfrom(2048)`
IP + portnumber

`modifiedMessage = message.decode().upper()`

send upper case string back to this client → `serverSocket.sendto(modifiedMessage.encode(), clientAddress)`

Example app: UDP client

Python UDPClient

We did not
specify the client
port number

either the IP address (e.g.,
“128.138.32.126”) or the hostname
(e.g., “cis.poly.edu”)

Create the client's socket

```
from socket import *  
serverName = 'hostname'  
serverPort = 12000
```

```
clientSocket = socket(AF_INET,  
                      SOCK_DGRAM)
```

get user keyboard
input

```
message = raw_input('Input lowercase sentence:')
```

Attach server name, port to
message; send into socket

```
clientSocket.sendto(message.encode(),  
                    (serverName, serverPort))
```

read reply characters from
socket into string

```
modifiedMessage, serverAddress =  
    clientSocket.recvfrom(2048)
```

print out received string
and close socket

```
print modifiedMessage.decode()  
clientSocket.close()
```


Socket programming with TCP

Client must **contact** server

- ❖ **server process must first be running**
- ❖ server must have created socket (door) that welcomes client's contact

Client contacts server by:

- ❖ Creating TCP socket, specifying IP address, port number of server process
- ❖ Client TCP establishes connection to server TCP

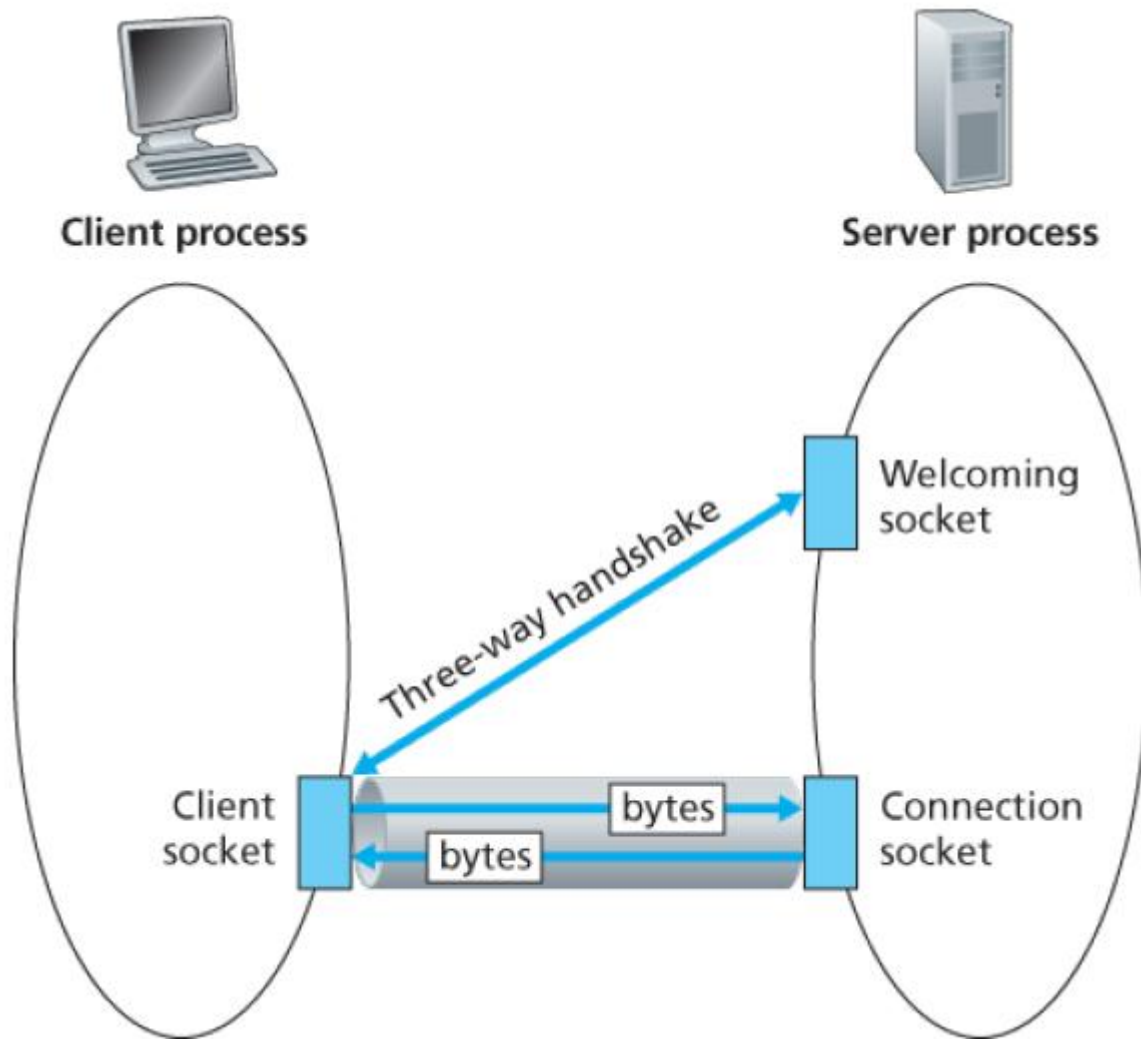
- ❖ when contacted by client, *server TCP creates new socket* for server process to communicate with that particular client
 - allows server to talk with multiple clients
 - source port numbers used to distinguish clients (more in Chap 3)

TCP socket is identified by (destination IP address, destination port number, source IP address, source port number)

Application viewpoint:

TCP provides reliable, in-order byte-stream transfer (“pipe”) between client and server

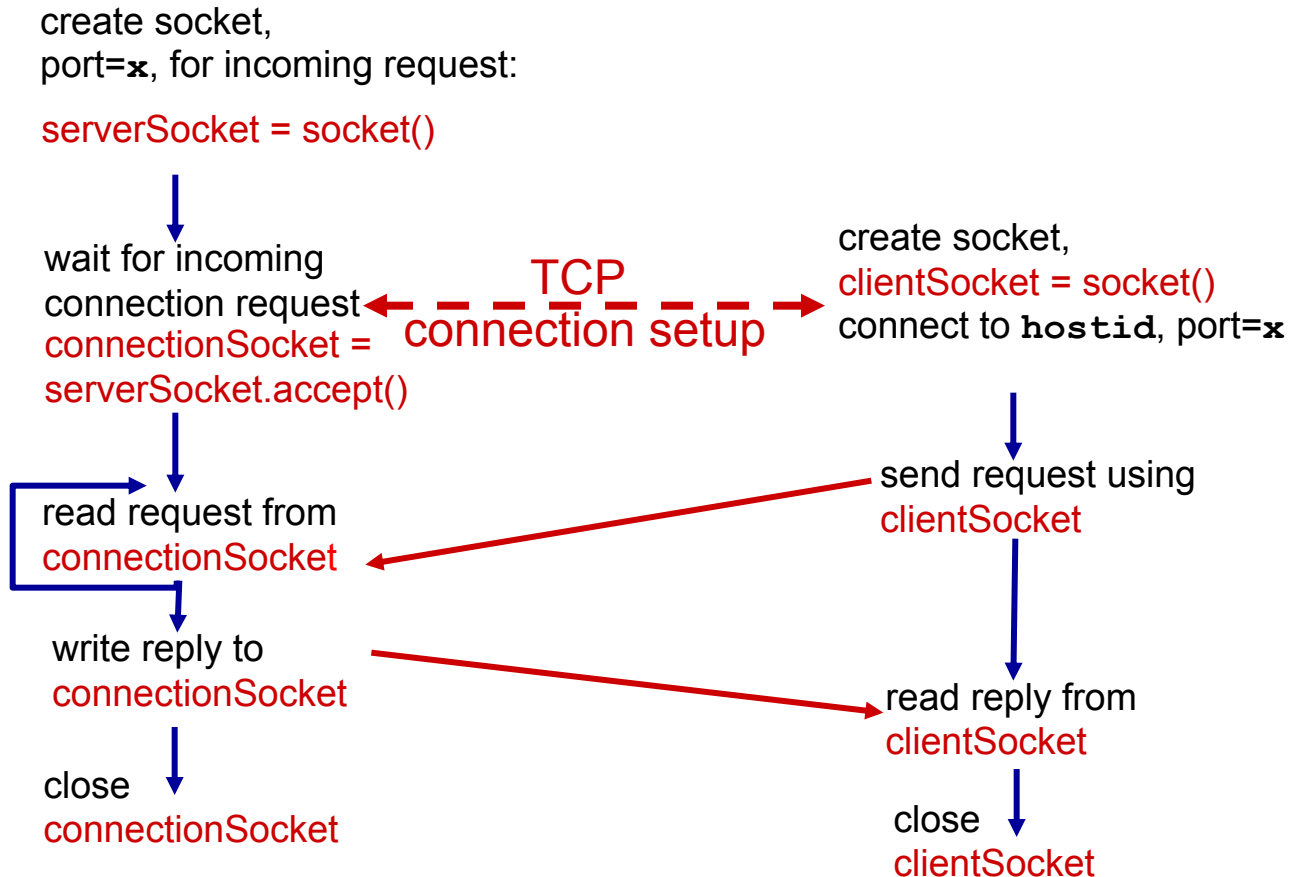
Socket programming with TCP



Client/server socket interaction: TCP

server (running on `hostid`)

client



Example app: TCP server

Python TCPServer

`connectionSocket` is identified by

- destination IP address and port number
- source IP address and port number

TCP socket

create TCP welcoming
socket

server begins listening for
incoming TCP requests

loop forever

server waits on `accept()`
for incoming requests, new
socket created on return

read bytes from socket (but
not address as in UDP)

close connection to this client
(but *not* welcoming socket)

```
from socket import *
```

```
serverPort = 12000
```

```
serverSocket = socket(AF_INET, SOCK_STREAM)
```

```
serverSocket.bind(('',serverPort))
```

```
serverSocket.listen(1)
```

```
print 'The server is ready to receive'
```

```
while True:
```

```
    connectionSocket, addr = serverSocket.accept()
```

```
    sentence = connectionSocket.recv(1024).decode()
```

```
    capitalizedSentence = sentence.upper()
```

```
    connectionSocket.send(capitalizedSentence.  
                           encode())
```

```
    connectionSocket.close()
```

Example app: TCP client

Python TCPCClient

```
from socket import *
```

```
serverName = 'servername'
```

```
serverPort = 12000
```

create TCP socket for server

```
→ clientSocket = socket(AF_INET, SOCK_STREAM)
```

```
clientSocket.connect((serverName,serverPort))
```

```
sentence = raw_input('Input lowercase sentence:')
```

No need to attach server
name, port

```
→ clientSocket.send(sentence.encode())
```

```
modifiedSentence = clientSocket.recv(1024)
```

```
print ('From Server:', modifiedSentence.decode())
```

```
clientSocket.close()
```

Chapter 2: summary

our study of network apps now complete!

- application architectures
 - client-server
 - P2P
- application service requirements:
 - reliability, bandwidth, delay
- Internet transport service model
 - connection-oriented, reliable: TCP
 - unreliable, datagrams: UDP
- specific protocols:
 - HTTP
 - SMTP, POP, IMAP
 - DNS
 - P2P: BitTorrent
- video streaming, CDNs
- socket programming:
TCP, UDP sockets

Chapter 2: summary

most importantly: learned about protocols!

- ❖ typical request/reply message exchange:
 - client requests info or service
 - server responds with data, status code
- ❖ message formats:
 - *headers*: fields giving info about data
 - *data*: info being communicated

important themes:

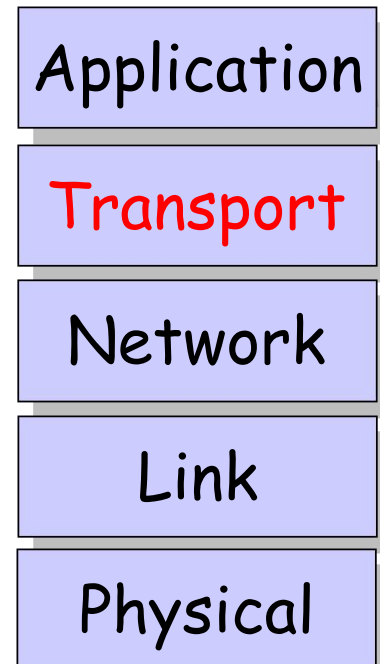
- centralized vs. distributed
- stateless vs. stateful
- reliable vs. unreliable message transfer

Chapter 3: Transport Layer

Our goals:

- ❖ understand principles behind transport layer services:
 - multiplexing, demultiplexing
 - reliable data transfer
 - flow control
 - congestion control

- ❖ learn about Internet transport layer protocols:
 - UDP: connectionless transport
 - TCP: connection-oriented reliable transport
 - TCP congestion control



Chapter 3 outline

3.1 transport-layer services

3.2 multiplexing and demultiplexing

3.3 connectionless transport: UDP

3.4 principles of reliable data transfer

3.5 connection-oriented transport: TCP

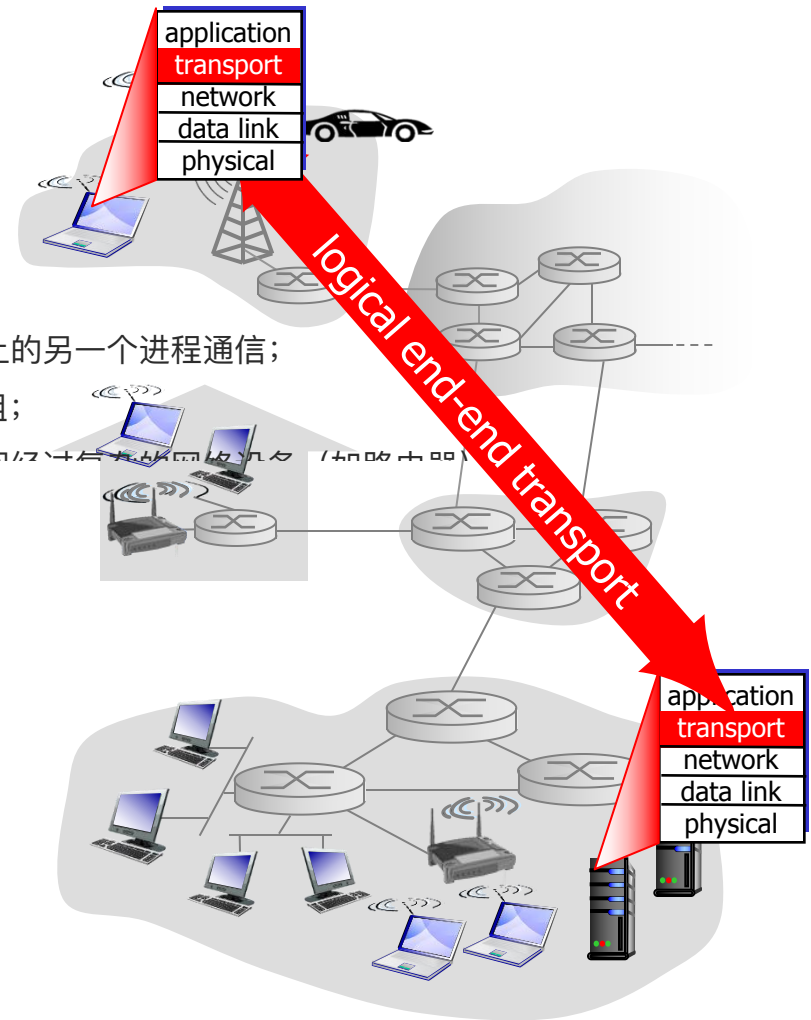
- segment structure
- reliable data transfer
- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control

Transport services and protocols

- ❖ Provide *logical communication* between app processes running on different hosts
- ❖ Transport protocols run in end systems
 - send side: breaks app messages into **segments**, passes to network layer
 - rcv side: **reassembles** segments into messages, passes to app layer
- ❖ more than one transport protocol available to apps
 - Internet: TCP and UDP

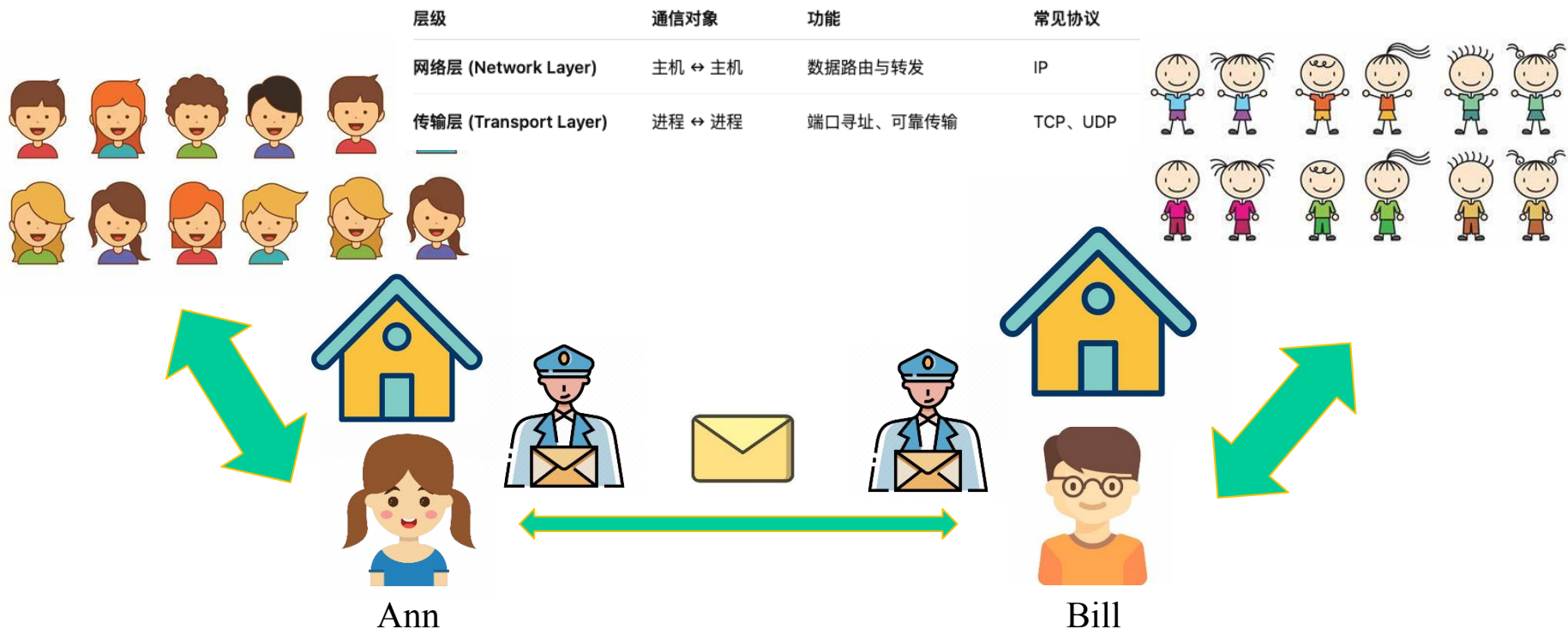


只有通信的两台计算机（例如客户端和服务端）才运行传输层协议；

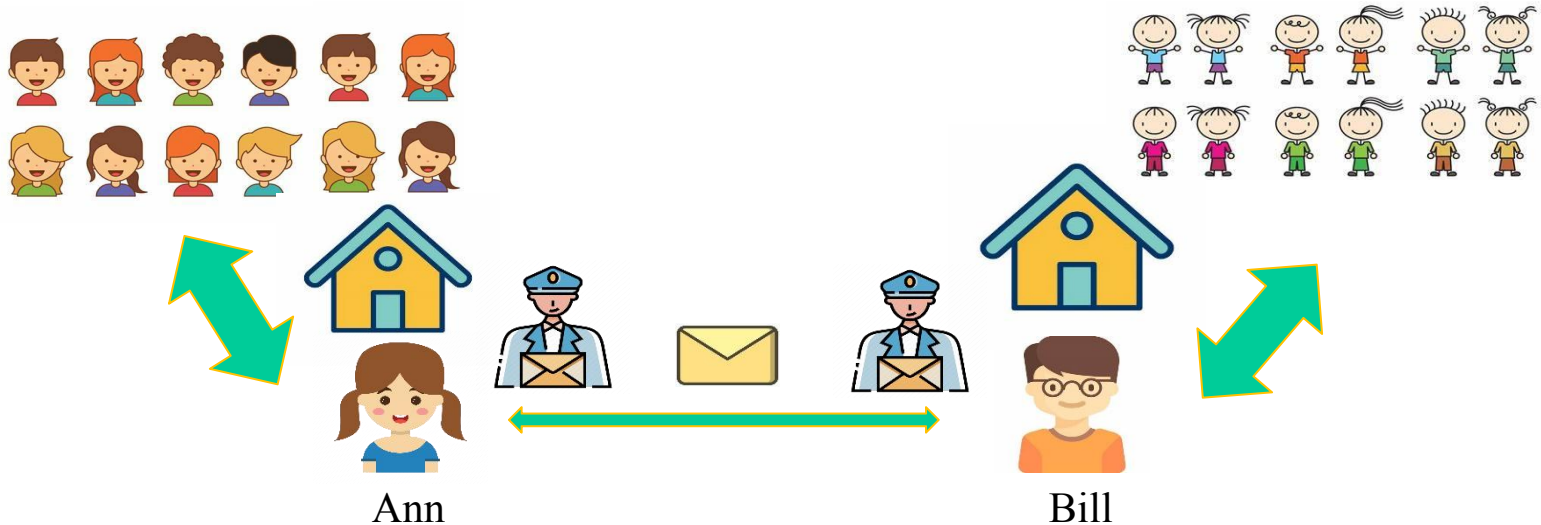
中间的路由器、交换机只负责转发数据包（它们只涉及网络层及以下的内容）。

Transport vs. network layer

- ❖ Network layer: logical communication between **hosts**
- ❖ Transport layer: logical communication between **processes**
 - relies on, enhances, network layer services



Transport vs. network layer



household analogy:

- ❖ hosts = houses
- ❖ processes = kids
- ❖ app messages = letters in envelopes
- ❖ **transport protocol = Ann and Bill**
- ❖ **network-layer protocol = postal service**

Susan and Harvey substitute for them and provide different delivery services?

Transport vs. network layer

- ❖ The services that a transport protocol can provide are often **constrained by** the service model of the underlying network-layer protocol. 传输层能提供的服务，往往受到网络层服务模型的限制。
 - delay or bandwidth guarantees
如果网络层（比如 IP 协议）不保证传输速度或带宽；
那么传输层协议也无法强制提供这些性能保证
- ❖ Certain services can be offered by a transport protocol **even when** the underlying network protocol **doesn't offer** the corresponding service at the network layer.
 - Reliable data transfer; security 传输层可以提供一些网络层没有提供的服务。

网络层（IP）是不可靠的 —— 它可能丢包、乱序、重复；

但传输层（如 TCP）能在上层提供：

可靠数据传输（reliable data transfer）

安全传输（security）

Internet transport-layer protocols

Network layer: Internet protocol (IP) is a **best effort delivery service, unreliable**

IP 不保证:

- 不丢包;

UDP: unreliable, unordered delivery;

- ❖ no-frills extension
- ❖ process-to-process data delivery and error checking

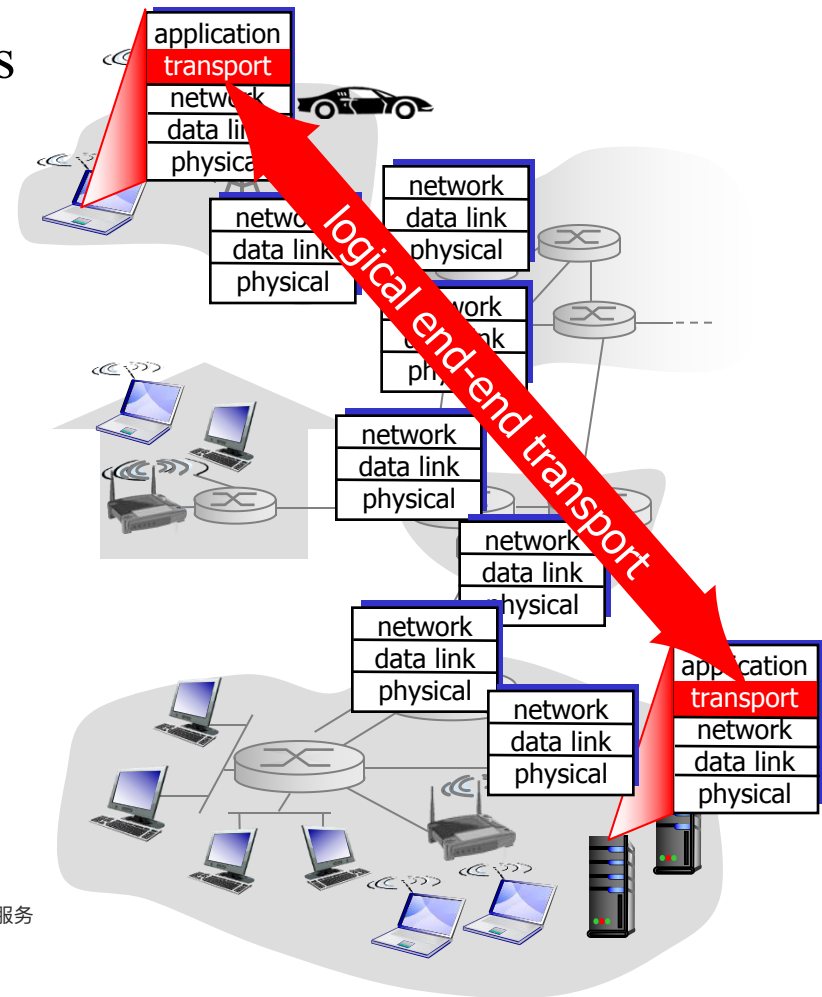
TCP : reliable, in-order delivery

- ❖ congestion control
- ❖ flow control
- ❖ connection setup

Services not available:

- ❖ delay guarantees
- ❖ bandwidth guarantees

无论是 TCP 还是 UDP, 都有一些不能提供的服务



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- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control

Multiplexing/demultiplexing

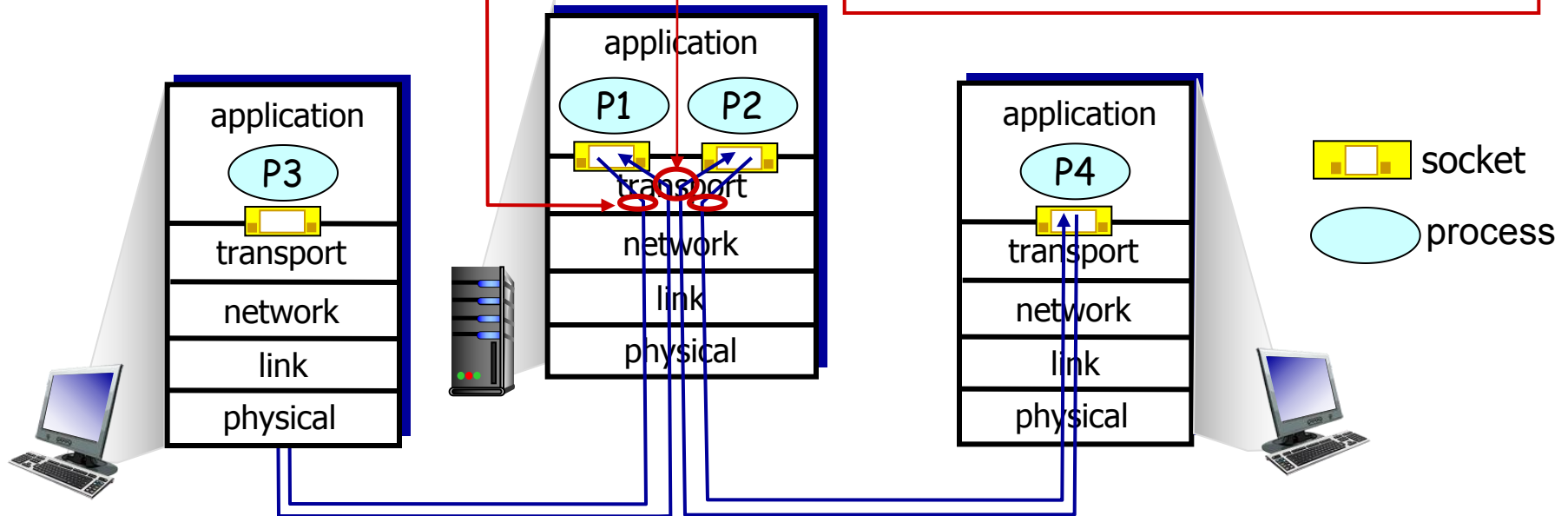
Multiplexing and demultiplexing: extending the host-to-host delivery service to a process-to-process delivery service for applications running on the hosts.

Multiplexing at sender:

handle data from multiple sockets, add **transport header** (later used for demultiplexing)

Demultiplexing at receiver:

use **header info** to deliver received segments to correct socket



Ann and Bill example?

◆1. Multiplexing (多路复用)

在发送端 (sender) :

- 一台主机上可能有很多不同的应用进程 (process), 例如:
 - 浏览器 (HTTP)
 - 邮件客户端 (SMTP)
 - 聊天程序 (Messenger)
- 每个进程都通过一个 **socket (套接字)** 与传输层交互。

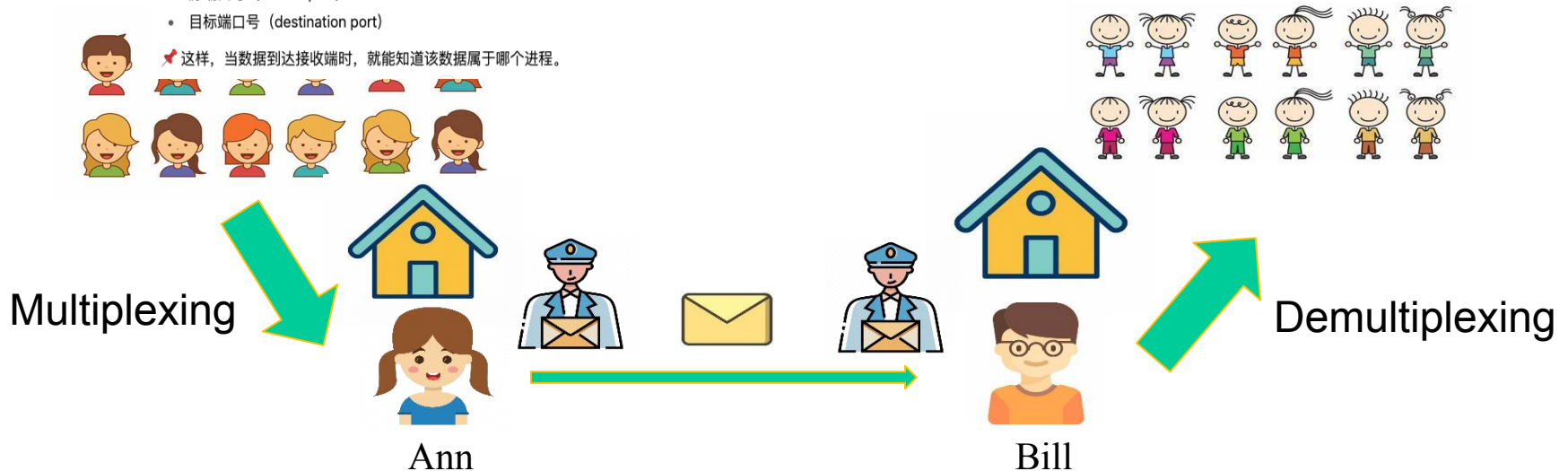
传输层要做的就是:

把来自多个 socket 的数据**复用 (multiplex)**到同一个网络连接中, 并为每个数据块添加一个 **传输层头部 (transport header)**。

这些头部里包含:

- 源端口号 (source port)
- 目标端口号 (destination port)

✚ 这样, 当数据到达接收端时, 就能知道该数据属于哪个进程。



◆2. Demultiplexing (多路分解)

在接收端 (receiver) :

- 主机从网络层接收到很多段 (segments)。
- 每个段的头部都包含源端口号和目的端口号。
- 传输层会根据这些信息:

把数据“分解 (demultiplex)”到正确的 **socket** 中, 交给对应的应用进程。

例如:

- 目标端口号 = 80 → 属于 Web 服务器进程;
- 目标端口号 = 25 → 属于邮件服务器进程。

How demultiplexing works?

- Each child must have an identifier (e.g., name, ID)

How demultiplexing works

Multiplexing/demultiplexing:

- ❖ sockets have **unique identifiers**
- ❖ each segment have special fields that indicate the socket to which the segment is to be delivered.

一台主机可能有多个应用进程（每个进程通过一个 socket 通信）。

传输层要从这些进程中收集数据，为每个数据段加上头部信息（source port、destination port 等），然后发送到网络层。

Sending host: Host uses *IP addresses* & *port numbers* to direct segment to appropriate socket

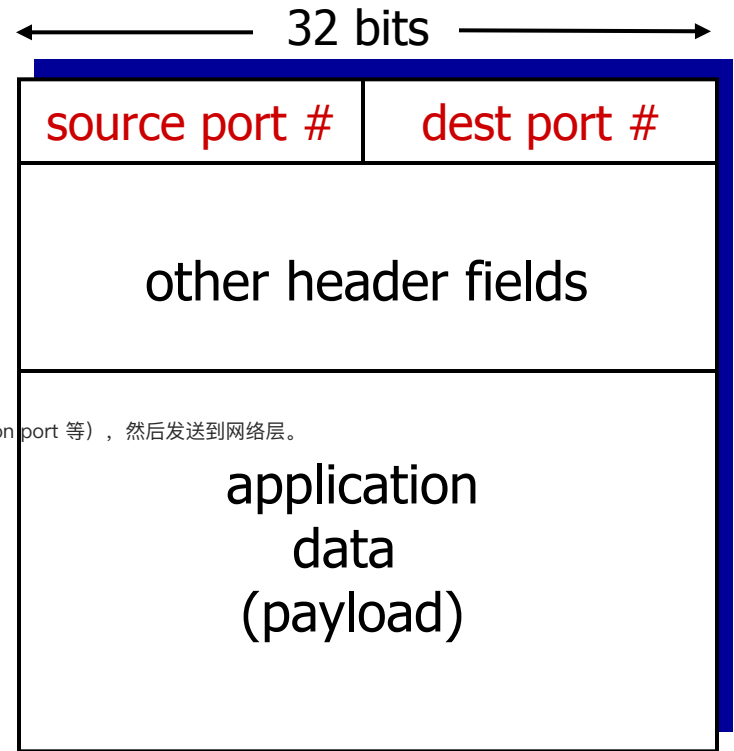
Receiving host: Host receives IP datagrams from network layer

- ❖ each datagram has **source IP address, destination IP address**
- ❖ each datagram carries one **transport-layer segment**

网络层收到数据报，交给传输层；

传输层根据目的 IP 和目的端口号，确定该数据属于哪个 socket；

将数据交给相应的应用进程。



TCP/UDP segment format

Port number: a 16-bit number, ranging from 0 to 65535:

- well-known port numbers: 0-1023

Overview

- ❖ UDP: Connectionless demux
- ❖ TCP: Connection-oriented demux

UDP: Connectionless demux

- ❖ *recall*: created socket has local port #:

```
clientSocket = socket(AF_INET, SOCK_DGRAM)
```

← Automatically assigns a port number

```
clientSocket.bind(('', 19157))
```

← Optional; at the server side, usually assign port number

- ❖ *recall*: when creating datagram to send into UDP socket, **must** specify destination IP address and destination port #

```
clientSocket.sendto(message.encode(), (serverName, serverPort))
```

UDP 的 socket 由一个“二元组 (two-tuple)”唯一确定:

UDP socket is fully identified by a **two-tuple** consisting of a destination IP address and a destination port number.

when host receives UDP segment:

- ❖ checks destination port # in segment
- ❖ directs UDP segment to socket with that port #



IP datagrams with *same dest. port #*, but different source IP addresses and/or source port numbers will be **directed to same socket** at destination

当主机收到一个 UDP 数据报时:

- 1 检查报文段头部的“目的端口号”;
- 2 查找系统中绑定了这个端口的 socket;

注意: UDP 不区分源 IP 或源端口。

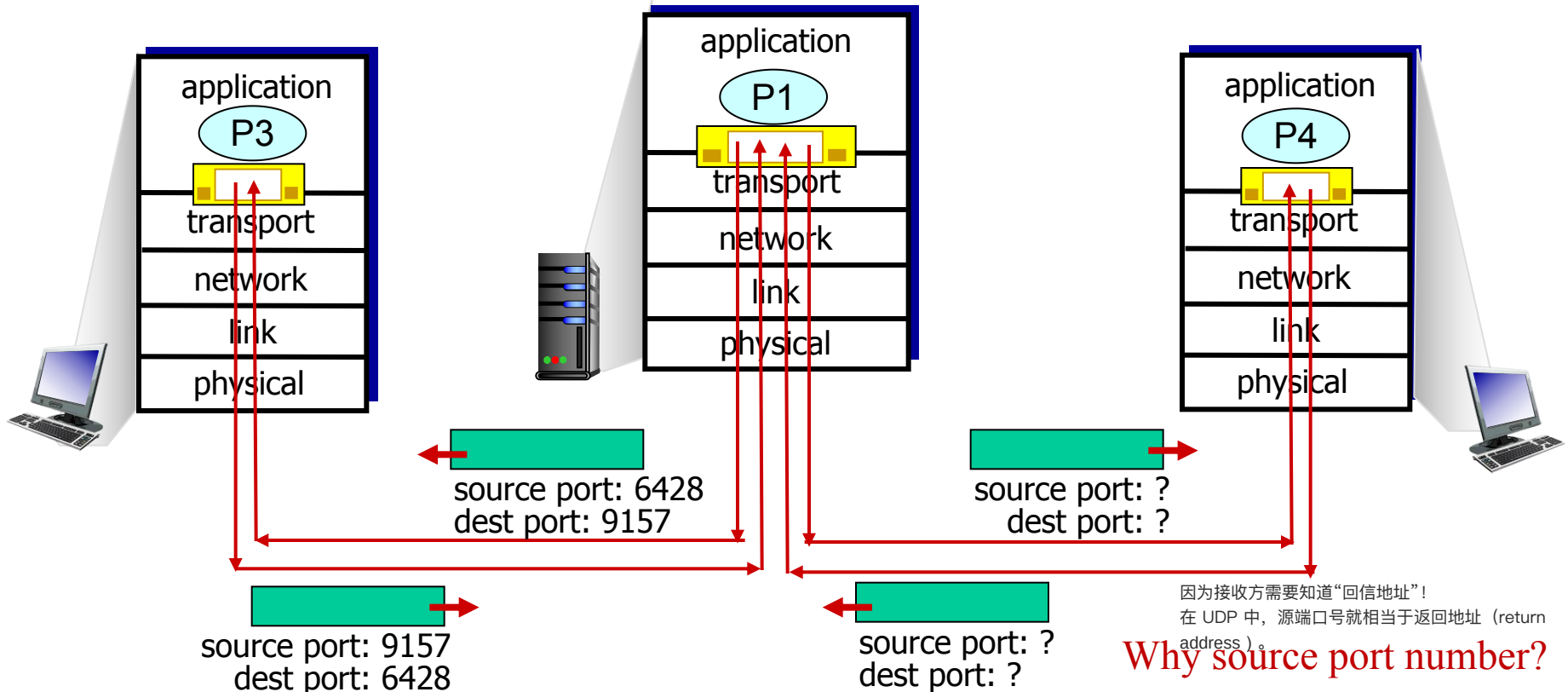
所以不同来源 (不同 IP、不同端口) 的数据, 只要目标端口号一样, 都会发给同一个 socket。

Connectionless demux: example

```
mysocket2 =  
    socket(AF_INET,  
          SOCK_DGRAM)  
mysocket2.bind  
    ((' ', 9157))
```

```
serversocket =  
    socket(AF_INET,  
          SOCK_DGRAM)  
serversocket.bind  
    ((' ', 6428))
```

```
mysocket1 =  
    socket(AF_INET,  
          SOCK_DGRAM)  
mysocket1.bind  
    ((' ', 5775))
```



因为接收方需要知道“回信地址”！
在 UDP 中，源端口号就相当于返回地址（return address）。

Why source port number?
❖ As the “return address”

Overview

- ❖ UDP: Connectionless demux
- ❖ TCP: Connection-oriented demux

TCP: Connection-oriented demux

- ❖ Server **creates a welcome socket** with port no.12000
`serversocket = socket(AF_INET, SOCK_STREAM)`
`serversocket.bind((' ', 12000))`
- ❖ Client **connects to the server**, the request is a TCP segment with a flag bit = 1

客户端调用 connect();

系统自动生成一个 TCP 段 (segment)，其中有一个 SYN 标志位 (flag bit = 1)；

服务器接收到后会响应 SYN+ACK，最终三次握手建立连接。

- `clientsocket = socket(AF_INET, SOCK_STREAM)`
`clientsocket.connect((ServerName, 12000))`
- ❖ Server **creates a new socket** to accept the connection
`connectionsocket, addr = serversocket.accept()`

The server may maintain TCP connections with multiple clients, each has a **different connectionsocket**. How to demux?

Connection-oriented demux

- ❖ TCP socket identified by 4-tuple:
 - source IP address, source port number, dest IP address, dest port number
- ❖ Demux: receiver uses all four values to direct segment to appropriate socket
- ❖ Server host may support many simultaneous TCP sockets:
 - each socket identified by its own 4-tuple

即使所有客户端都连接到同一个端口（如 80），服务器仍能区分每个连接

Web servers have different sockets for each connecting client

- Both the initial connection-establishment segments and the segments carrying HTTP requests will have destination port 80.
- non-persistent HTTP will have different socket for each request

Connection-oriented demux: example

是的，服务器可以在同一个端口上拥有多个 socket。

端口号 (port number):

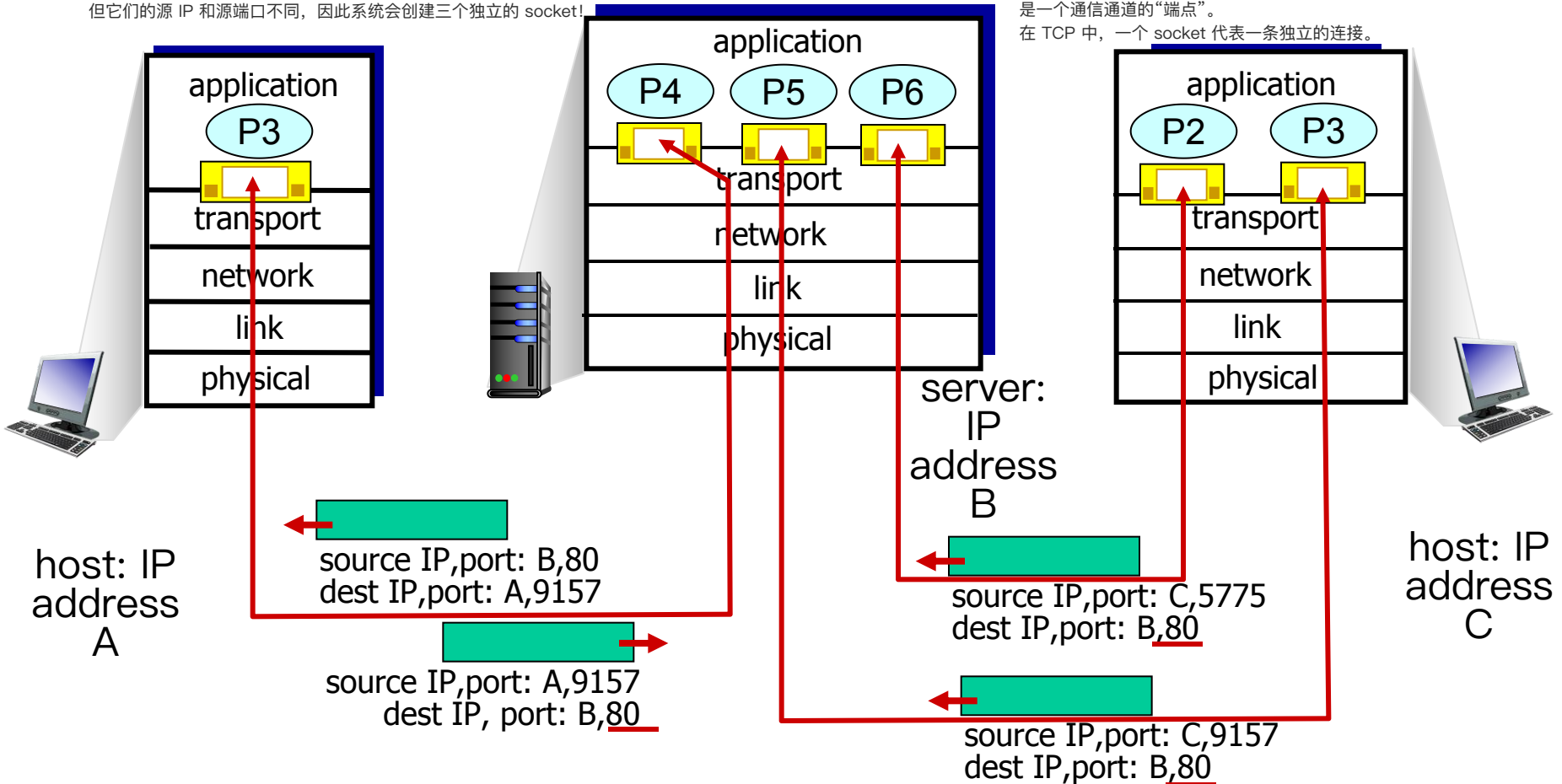
就像“门牌号”一样，是进入服务器通信的“入口”。

虽然三个连接的目标端口（服务器端口）都是 80，
但它们的源 IP 和源端口不同，因此系统会创建三个独立的 socket！

socket (套接字):

是一个通信通道的“端点”。

在 TCP 中，一个 socket 代表一条独立的连接。



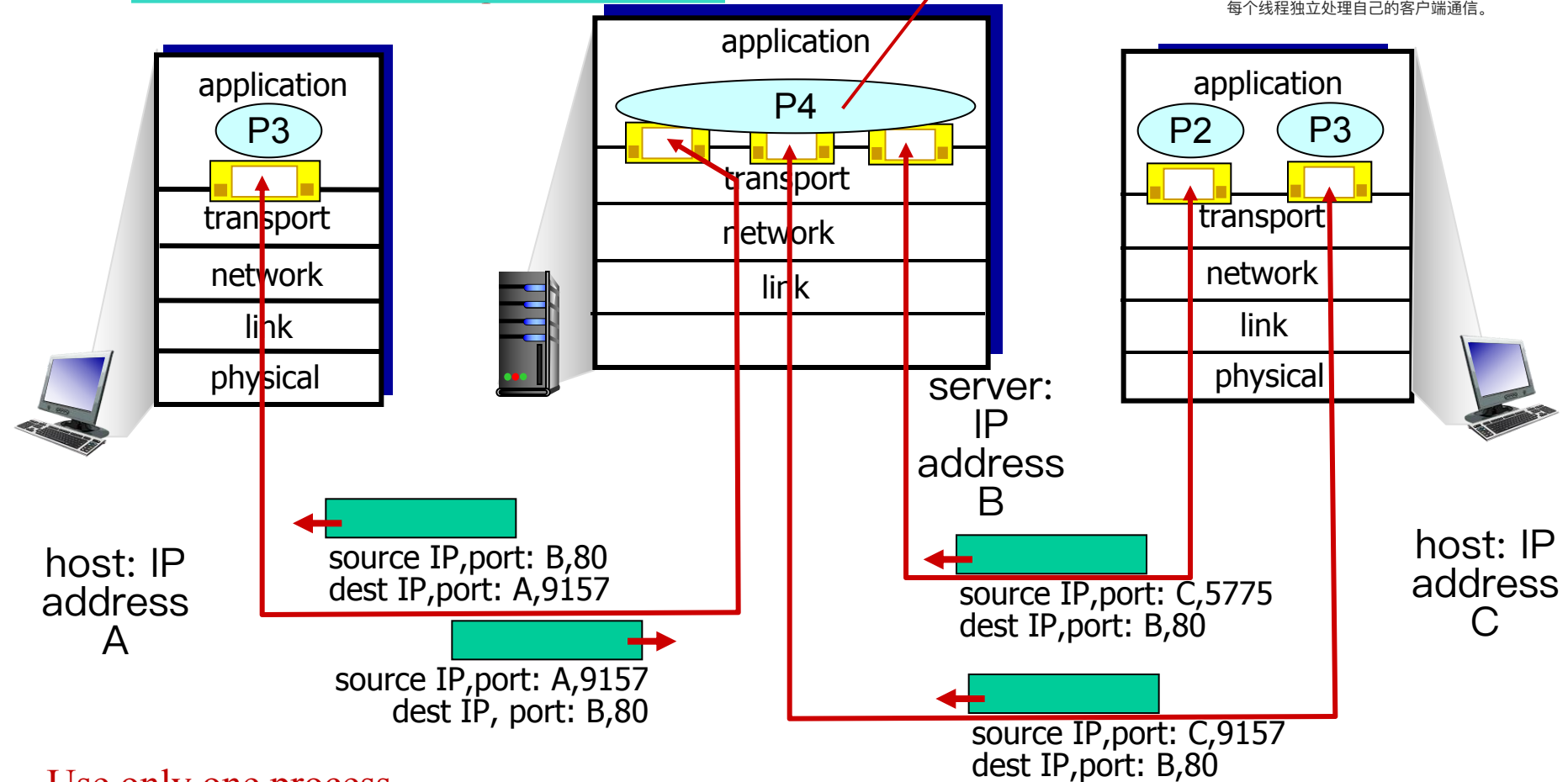
three segments, all destined to IP address: B, dest port: 80
are demultiplexed to *different sockets*

Connection-oriented demux: example

There is not always a one-to-one correspondence between connection sockets and processes

threaded server

服务器只运行一个主进程 (P4) ;
每当有新的客户端连接时:
服务器创建一个新的 线程 (thread);
这个线程绑定到该客户端的 socket;
每个线程独立处理自己的客户端通信。



Use only one process

- ❖ Create a new thread with a new connection socket for each new client connection.
- ❖ A thread can be viewed as a lightweight subprocess.

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3.2 multiplexing and demultiplexing

3.3 connectionless transport: UDP

3.4 principles of reliable data transfer

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- segment structure
- reliable data transfer
- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control

UDP: User Datagram Protocol

只做最基本的数据传输，其他不保证

- ❖ “No frills,” “bare bones” Internet transport protocol
 - Multiplexing/demultiplexing; light error checking
- ❖ “Best effort” service, UDP segments may be:
 - Lost, delivered out-of-order to app
- ❖ *connectionless*:
 - No handshaking between UDP sender, receiver
 - Each UDP segment handled independently of others
 - No congestion control

Advantage?

- No congestion control: Immediately pass the segment to network layer
- No connection-establish delay
- No connection state: server can support more clients
- Smaller packet overhead

Disadvantage?

- No congestion control: congestion, overflow, fairness
- Not reliable

UDP: User Datagram Protocol

❖ UDP is used in:

- streaming multimedia apps (loss tolerant, rate sensitive)
- DNS

❖ reliable transfer over UDP:

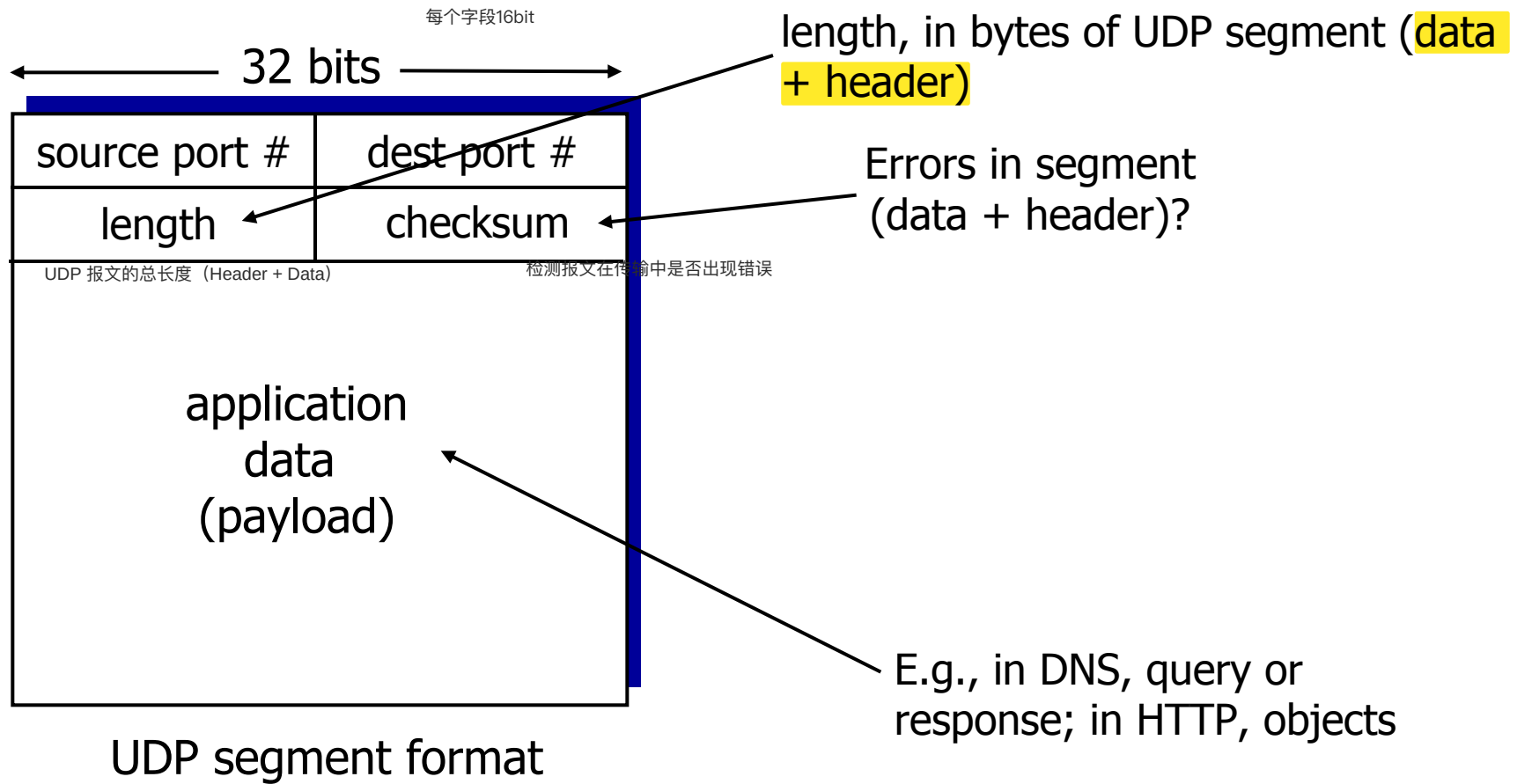
- add reliability at application layer
- application-specific error recovery!

在应用层增加 ACK（确认包）；

通过编号、超时重传机制实现可靠性；

甚至可在 UDP 上实现“自定义的可靠协议”（例如 QUIC 协议）。

UDP: segment header



UDP checksum

Goal: detect “errors” (e.g., flipped bits) in transmitted segment (from source to destination)

Sender:

- ❖ treat segment contents, including header fields, as sequence of 16-bit integers 长度不是偶数, 就在末尾补上0 把所有的16位数相加, 按位取
- ❖ checksum: **1s complement of the sum of segment contents**
- ❖ sender puts checksum value into UDP checksum field

把 checksum 填入 UDP header 的 checksum 字段。

Receiver:

- ❖ check the sum of the segment 包括sender计算出来的checksum
 - All bits **are equal to 1 - no error detected.** *But maybe errors nonetheless? More later*
 - Otherwise: error detected

Internet checksum: example

At the sender side, **determine the check sum** of the following three 16-bit words

0110011001100000
0101010101010101
1000111100001100

The sum of first two of these 16-bit words is

0110011001100000
0101010101010101

1011101110110101

Adding the third word to the above sum gives

1011101110110101
1000111100001100

10100101011000001
 **Wrap around**
0100101011000010

Note: when adding numbers, a carryout from the most significant bit needs to be added to the result

The 1s complement is obtained **by converting all the 0s to 1s and converting all the 1s to 0s.**

Checksum: 1011010100111101

Receiver Side?

Internet checksum: example

At the receiver side, all four 16-bit words are added, including the checksum:

- ❖ If no errors are introduced into the packet, then clearly the sum at the receiver will be 1111111111111111.
- ❖ If one of the bits is a 0, then we know that errors have been introduced into the packet.

Check the sum of the segment

- All bits are equal to 1 - **no error detected**. *But maybe errors nonetheless? More later*
- Otherwise: error detected

Internet checksum: example

Why UDP provides a checksum?

防止“底层没检测”导致错误数据直接送到应用层

- **no guarantee** that all the links provide error checking
- bit errors could be introduced when segments are **in memory**
- “functions placed at the lower levels may be redundant or of little value when compared to the cost of providing them at the higher level.”

在底层实现某些功能可能冗余或代价高，而在高层实现可能更灵活、更高效。

UDP 不重传、不纠错，只负责检测错误；真正的可靠性（要不要重发）由应用层决定。

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3.4 principles of reliable data transfer

3.5 connection-oriented transport: TCP

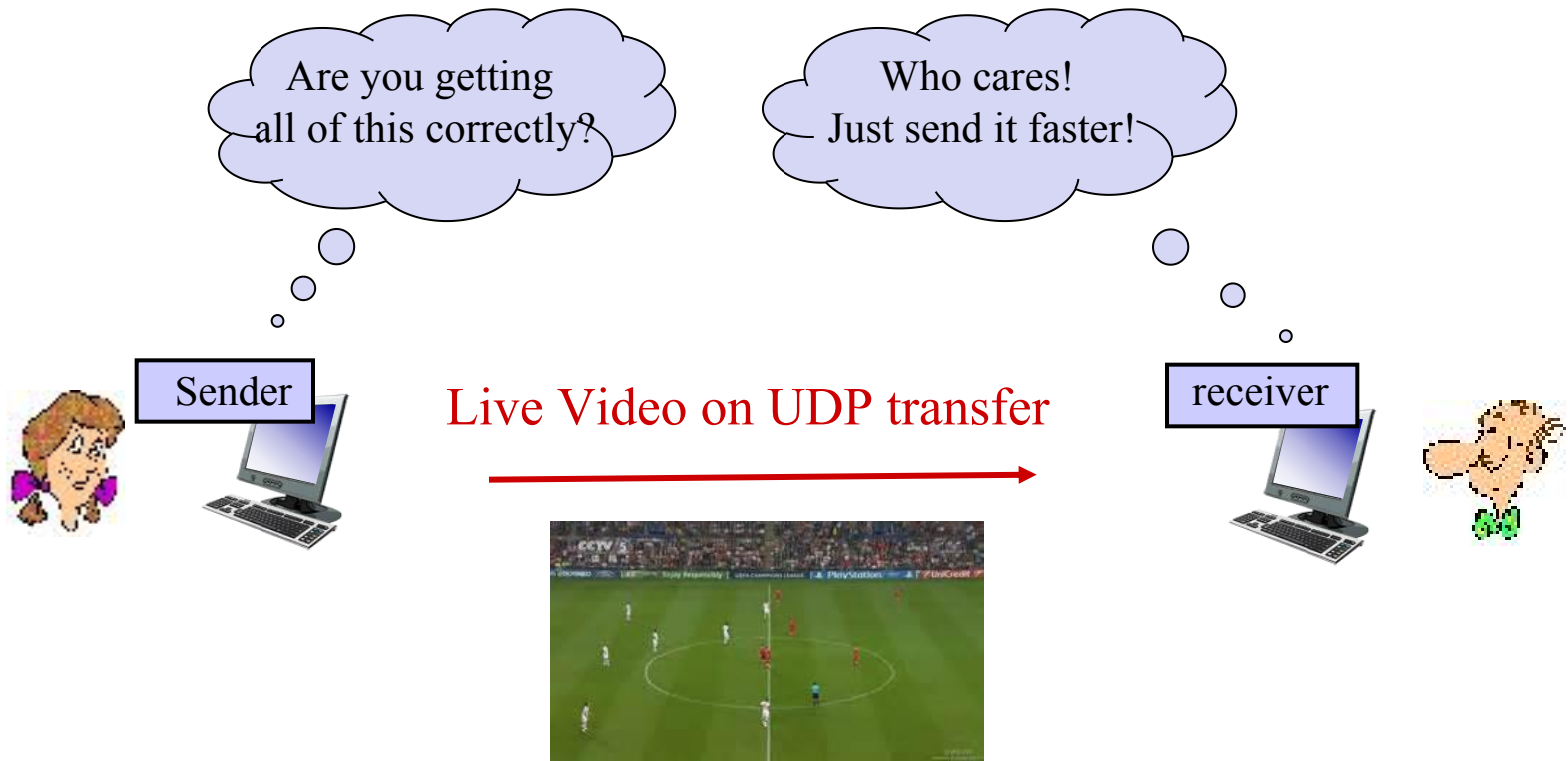
- segment structure
- reliable data transfer
- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control

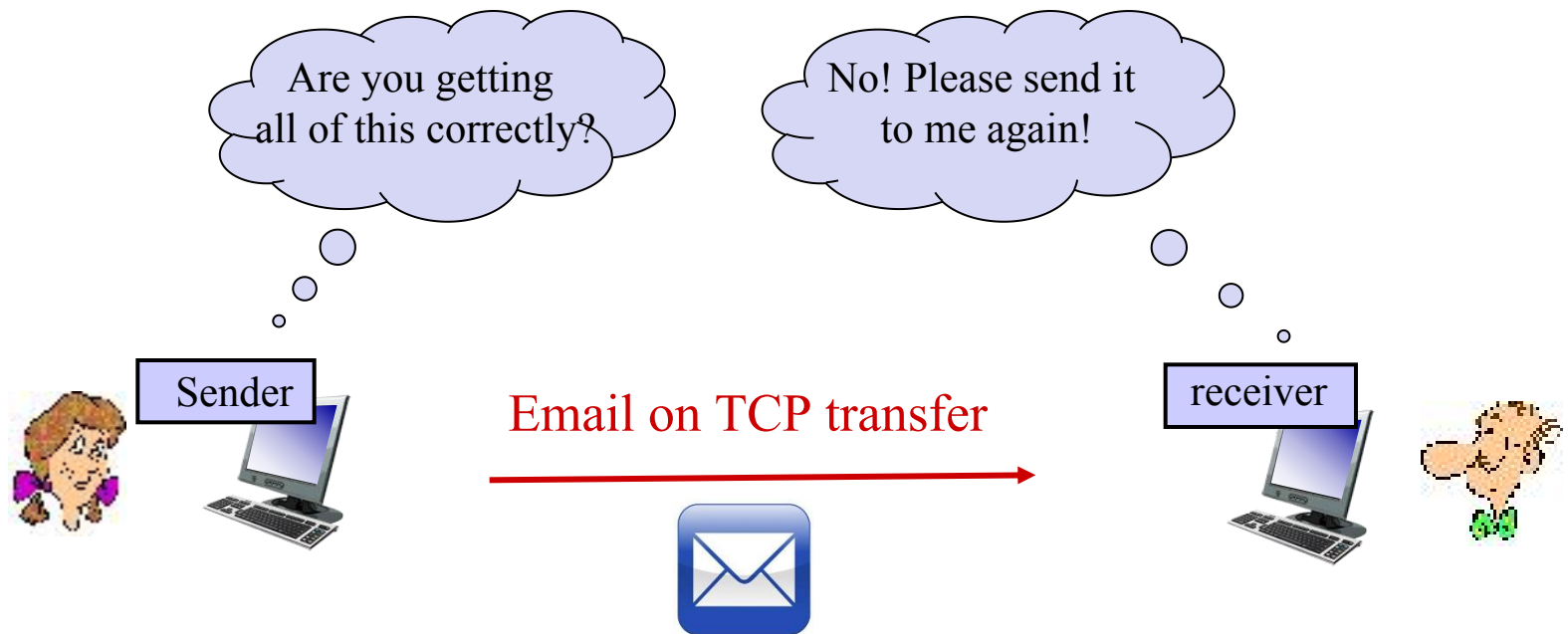
UDP Transfer

- ❖ UDP cannot guarantee reliable data transfer
- ❖ But, it's faster!



TCP Transfer

- ❖ TCP can guarantee reliable data transfer
- ❖ But, it's slower and more complex!



Reliable Data Transfer (rdt)

在不可靠信道上可靠传输数据的机制

- ❖ In top-10 list of important networking topics!
- ❖ Reliable data transfer **over unreliable channel**:
 - Bit flip, lost, out-of-order
 - In this section, assume unreliable channel not reorder packets



Nothing need to do!

Reliable Channel

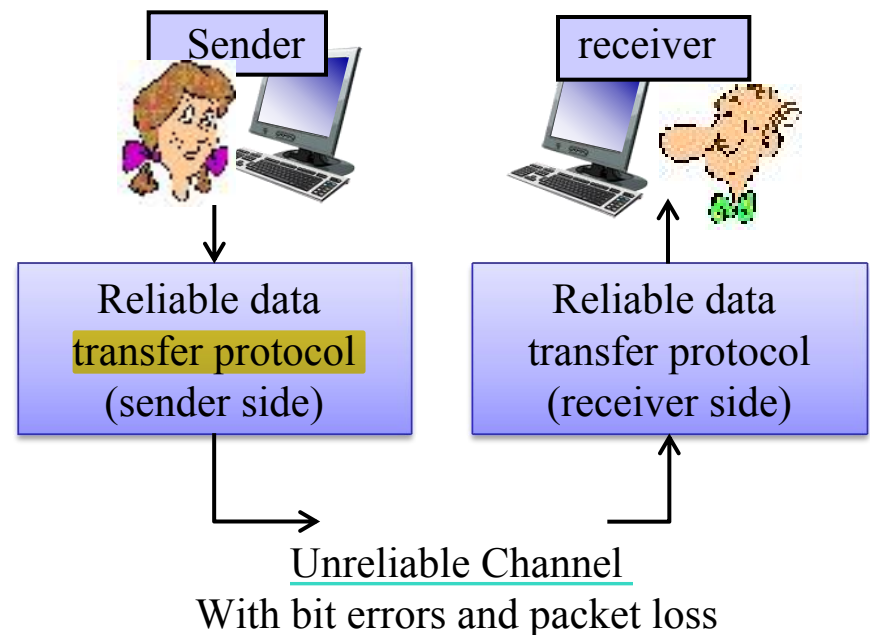
底层网络可能丢包、乱序、比特翻转;

但 RDT 协议必须保证:

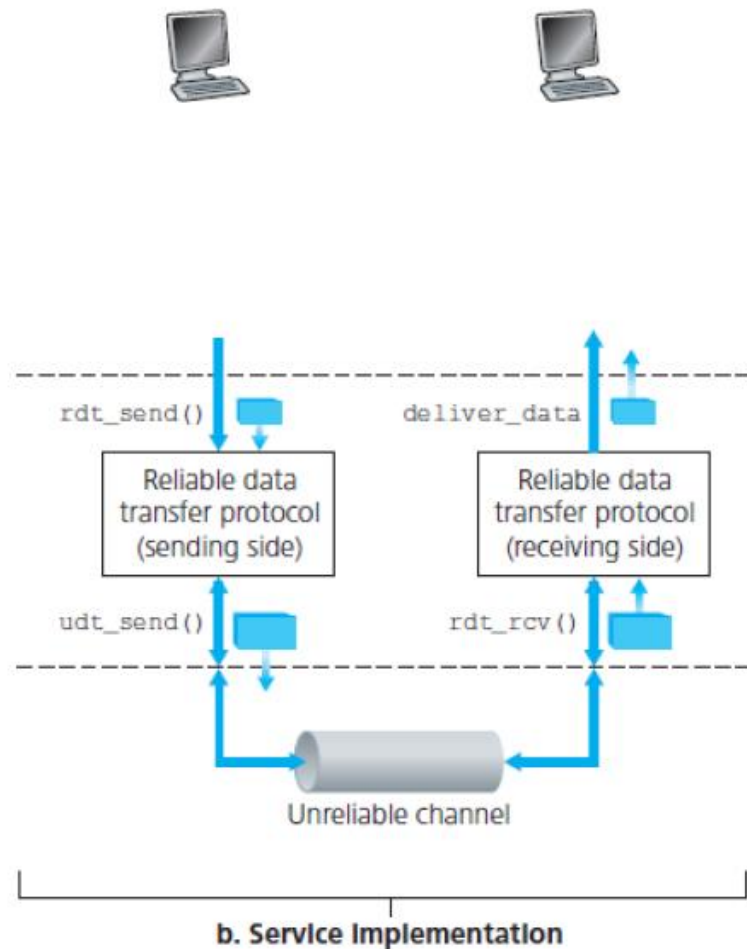
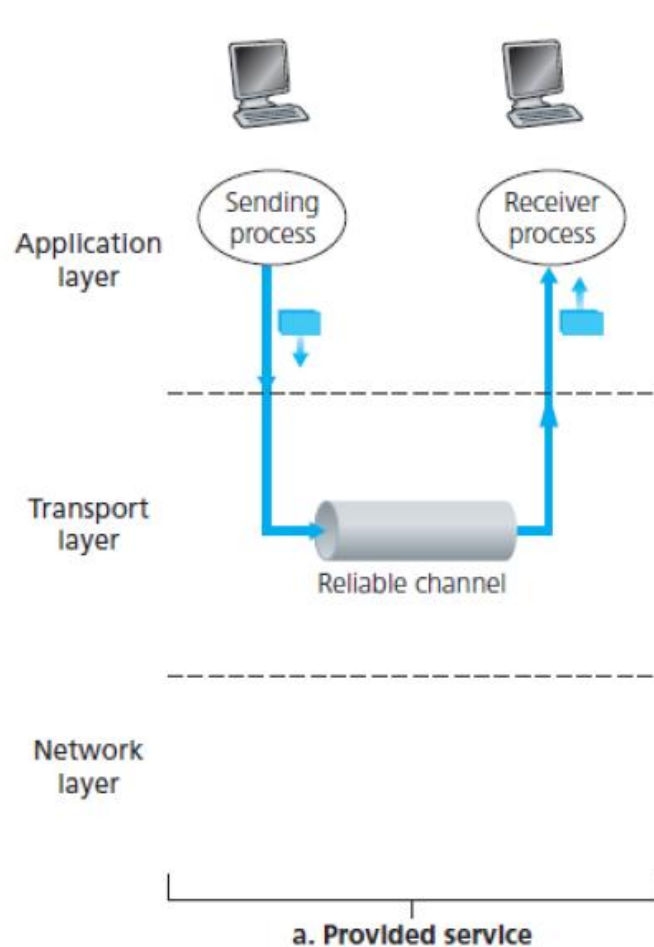
数据不丢失;

不重复;

按顺序交付给接收端。



Reliable Data Transfer (rdt)



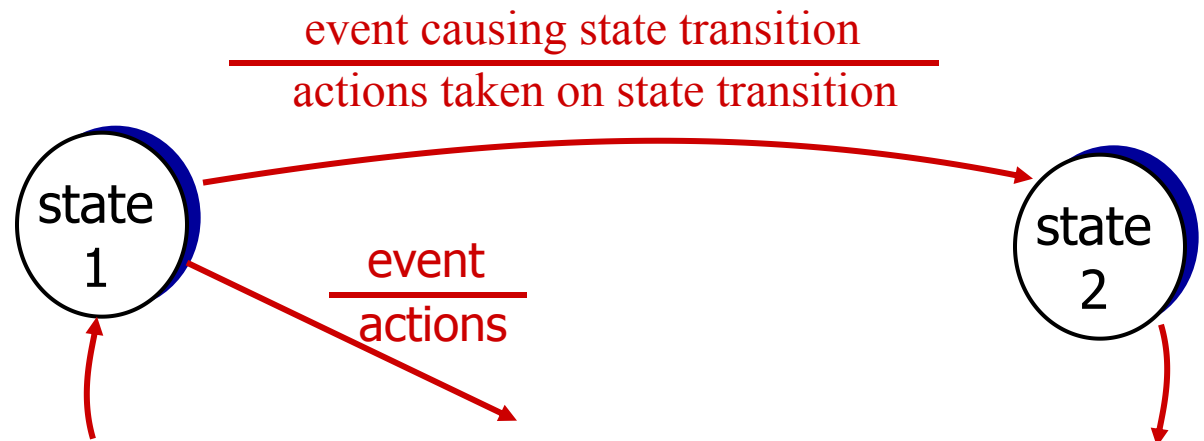
Reliable data transfer: getting started

We'll:

- ❖ Incrementally develop sender, receiver sides of reliable data transfer protocol (rdt)
- ❖ Consider only unidirectional data transfer
 - but control info will flow on both directions!
- ❖ Use finite state machines (FSM) to specify sender, receiver

控制信息（如确认ACK、否认NAK）仍然会双向流动

state: when in this “state”
next state uniquely
determined by next
event



Overview

Roadmap:

- ❖ Perfectly reliable channel: rdt1.0
- ❖ Channel with bit error:
 - bit error in packet: rdt 2.0
 - bit error in ACK: 2.1
 - NAK-free: 2.2
- ❖ Lossy channel: rdt 3.0

Summary of Techniques

- Checksum
- Sequence number
- ACK packets
- Retransmission
- Timeout

rdt1.0: reliable transfer over a reliable channel

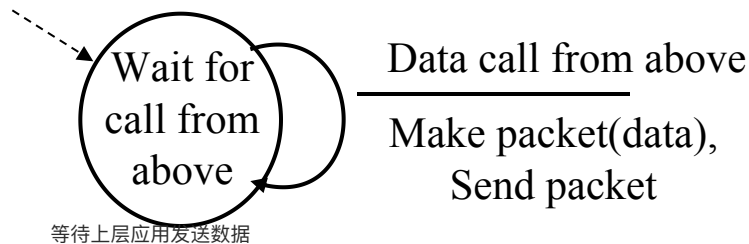
❖ Underlying channel **perfectly reliable**

- no bit errors
- no loss of packets

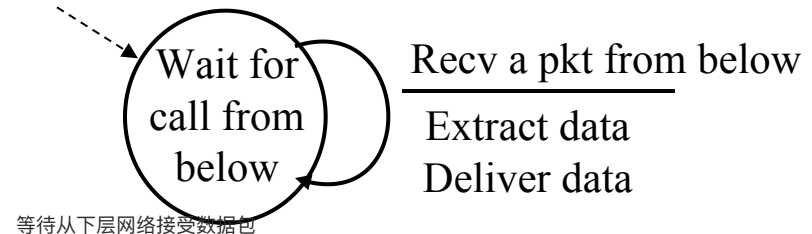


❖ **Rdt 1.0:**

- sender sends data into underlying channel
- receiver reads data from underlying channel
- Reliable channel, no need for feedback (no control message)

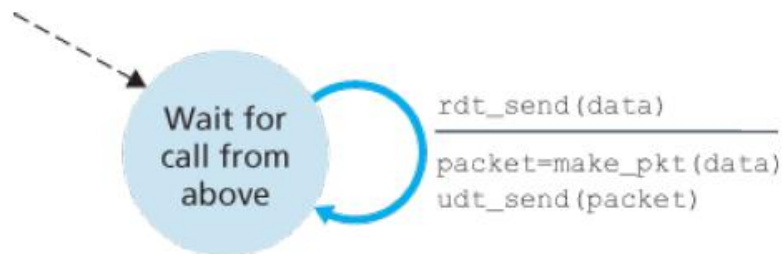
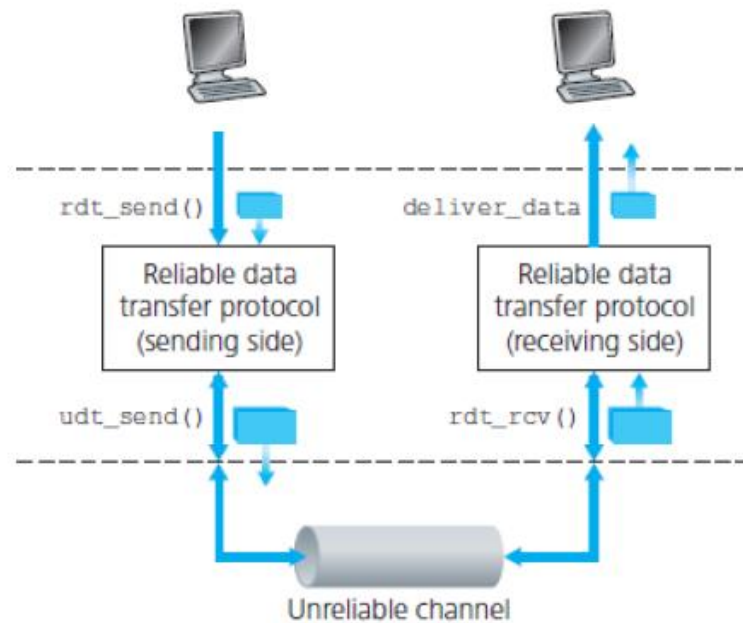


sender

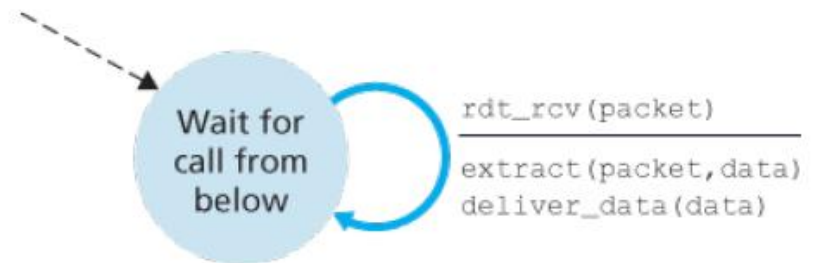


receiver

rdt1.0: reliable transfer over a reliable channel



sender

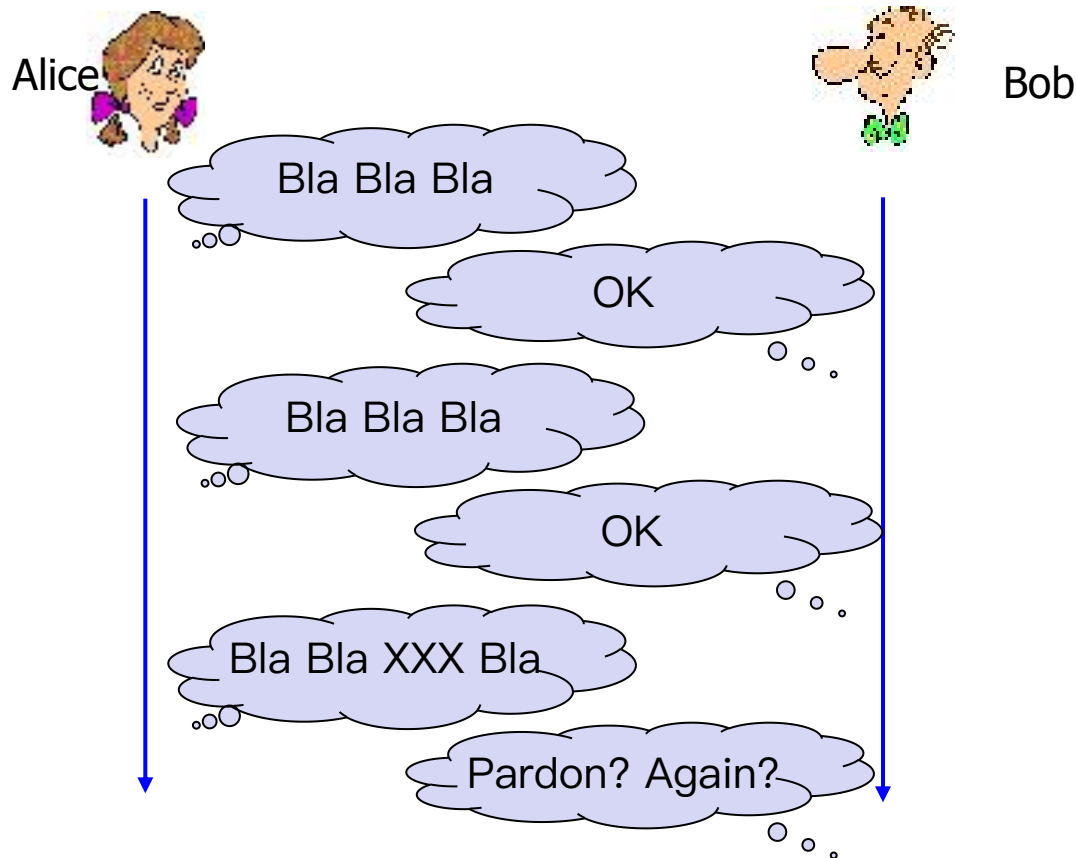


receiver

rdt2.0: channel with bit errors

- ❖ Underlying channel may **flip bits** ($0 \rightarrow 1$) in packet

How do humans recover from “errors” during conversation?



rdt2.0: channel with bit errors

- ❖ Underlying channel may **flip bits** ($0 \rightarrow 1$) in packet

How do humans recover from “errors” during conversation?

- ❖ **The question:** how to recover from errors?
 - *acknowledgements (ACKs)*: receiver explicitly tells sender that pkt received OK
 - *negative acknowledgements (NAKs)*: receiver explicitly tells sender that pkt had errors
 - sender retransmits pkt on receipt of NAK
- ❖ new mechanisms in **rdt2.0** (beyond **rdt1.0**):
 - error detection
 - receiver **feedback**: control msgs (ACK,NAK) rcvr->sender
 - retransmission

rdt2.0: channel with bit errors

❖ Key mechanisms:

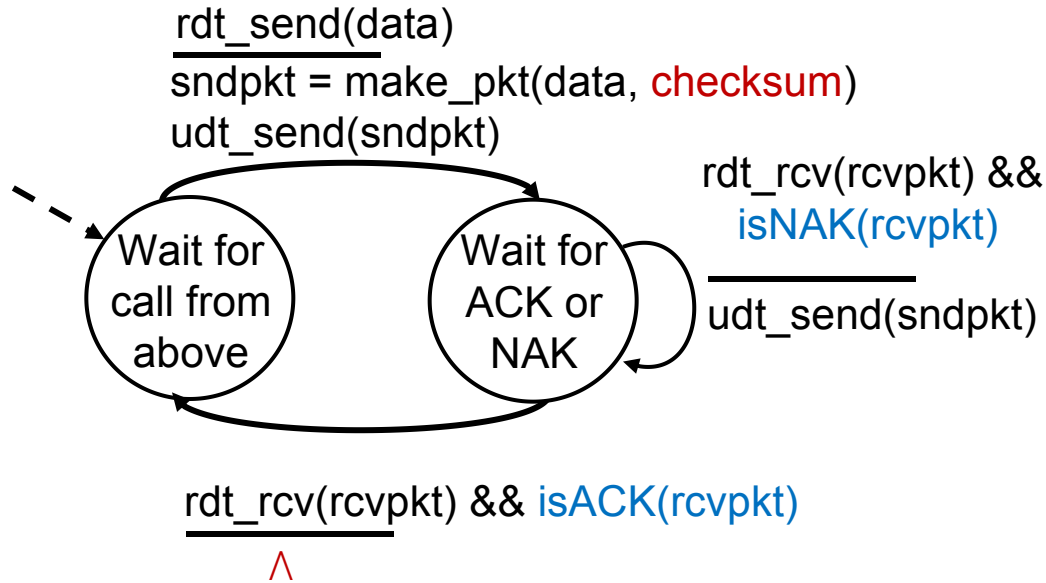
- **error detection**
- **feedback**: control msgs (ACK, NAK) from receiver to sender
- retransmission

❖ Error detection: checksum

❖ Feedback messages:

- *acknowledgements (ACKs)*: receiver explicitly tells sender that pkt received OK
- *negative acknowledgements (NAKs)*: receiver explicitly tells sender that pkt had errors
- sender retransmits pkt on receipt of NAK

rdt2.0: FSM specification

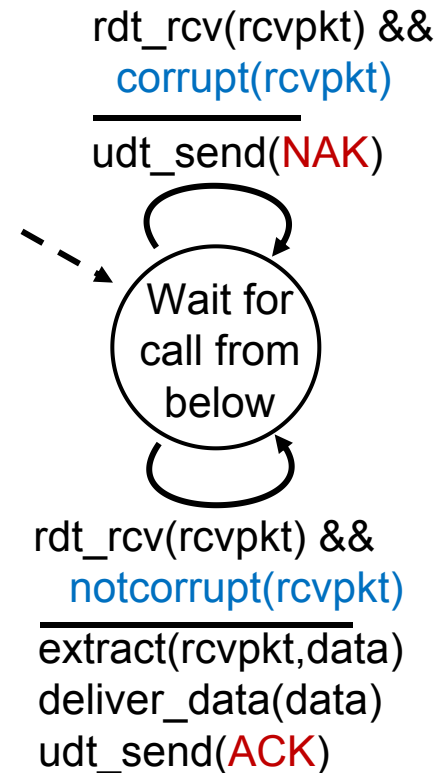


sender

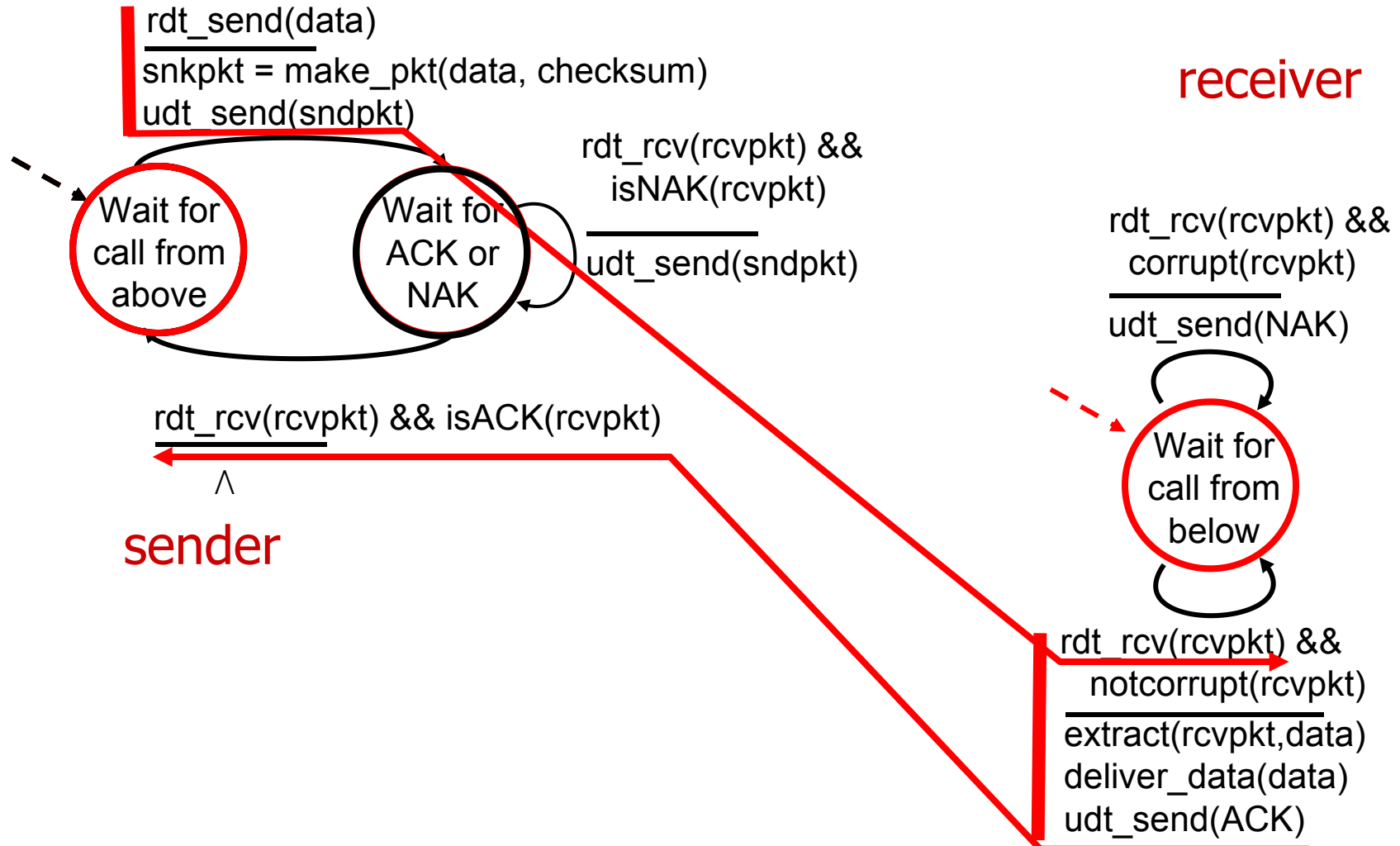
Stop and wait

Sender sends one packet,
then waits for receiver
response

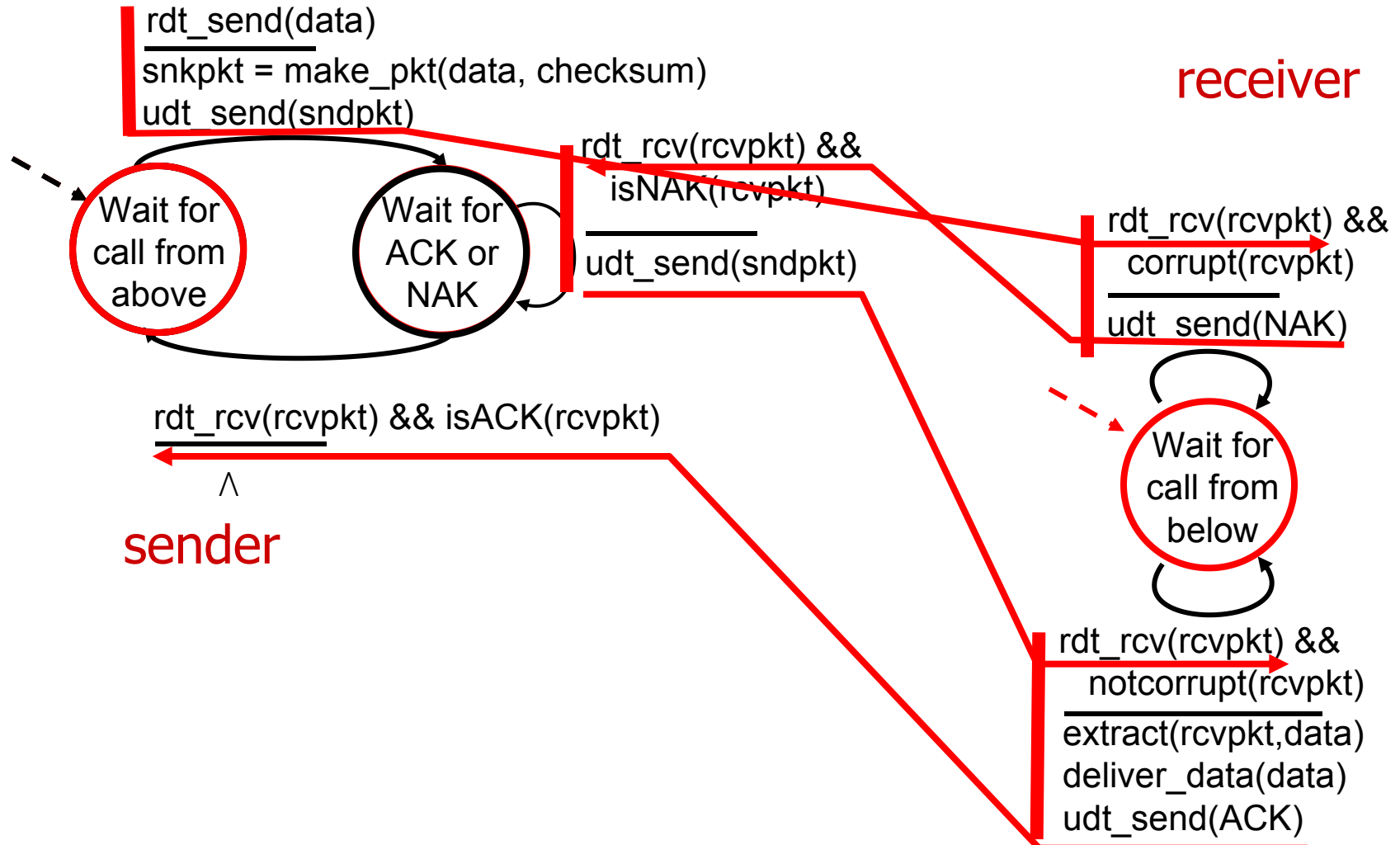
receiver



rdt2.0: operation with no errors



rdt2.0: error scenario



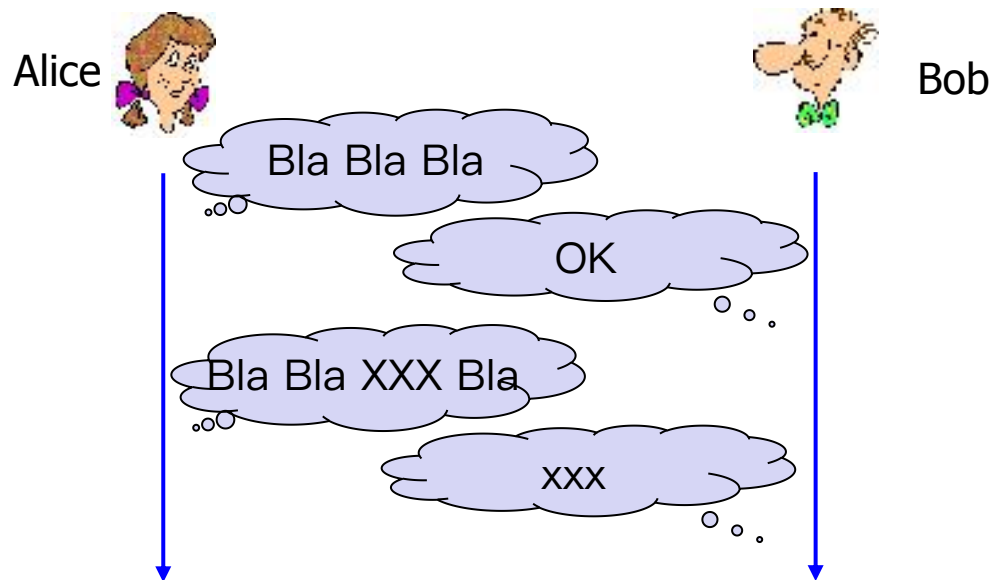
rdt2.0 has a fatal flaw!

The possibility that **ACK or NAK** packet could be corrupted:

- ❖ Checksum bits

Handling corrupted ACKs or NAKs:

- ❖ Option 1: “blabla...”, “OK”, “What did you say?”, “OK”
 - “What did you say?”, “What did you say?”, ...
- ❖ Option 2: add enough checksum to recover
- ❖ Option 3: when garbled ACK or NAK, retransmit



rdt2.0 has a fatal flaw!

The possibility that ACK or NAK packet could be corrupted:

- ❖ Checksum bits

Handling corrupted ACKs or NAKs:

- ❖ Option 1: “blabla...”, “OK”, “What did you say?”, “OK”
 - “What did you say?”, “What did you say?”, ... 要求对方重复说
- ❖ Option 2: add enough checksum to recover 成本高
- ❖ **Option 3: when garbled ACK or NAK, retransmit**

出现乱码ACK/NAK，直接重传

Problem: can't just retransmit: **new data or retransmission?**
possible duplicate

Handling duplicates:

- ❖ sender retransmits current pkt if ACK/NAK corrupted
- ❖ sender adds **sequence number** to each pkt
- ❖ receiver discards (doesn't deliver up) duplicate pkt

发送方在每个新数据包时切换序列号；

接收方通过序列号判断：

这是新包（正常处理）；

还是重复包（丢弃）。

只需要两个序列号（0 和 1）就够了。
发送方一次只会发送一个未确认的包。

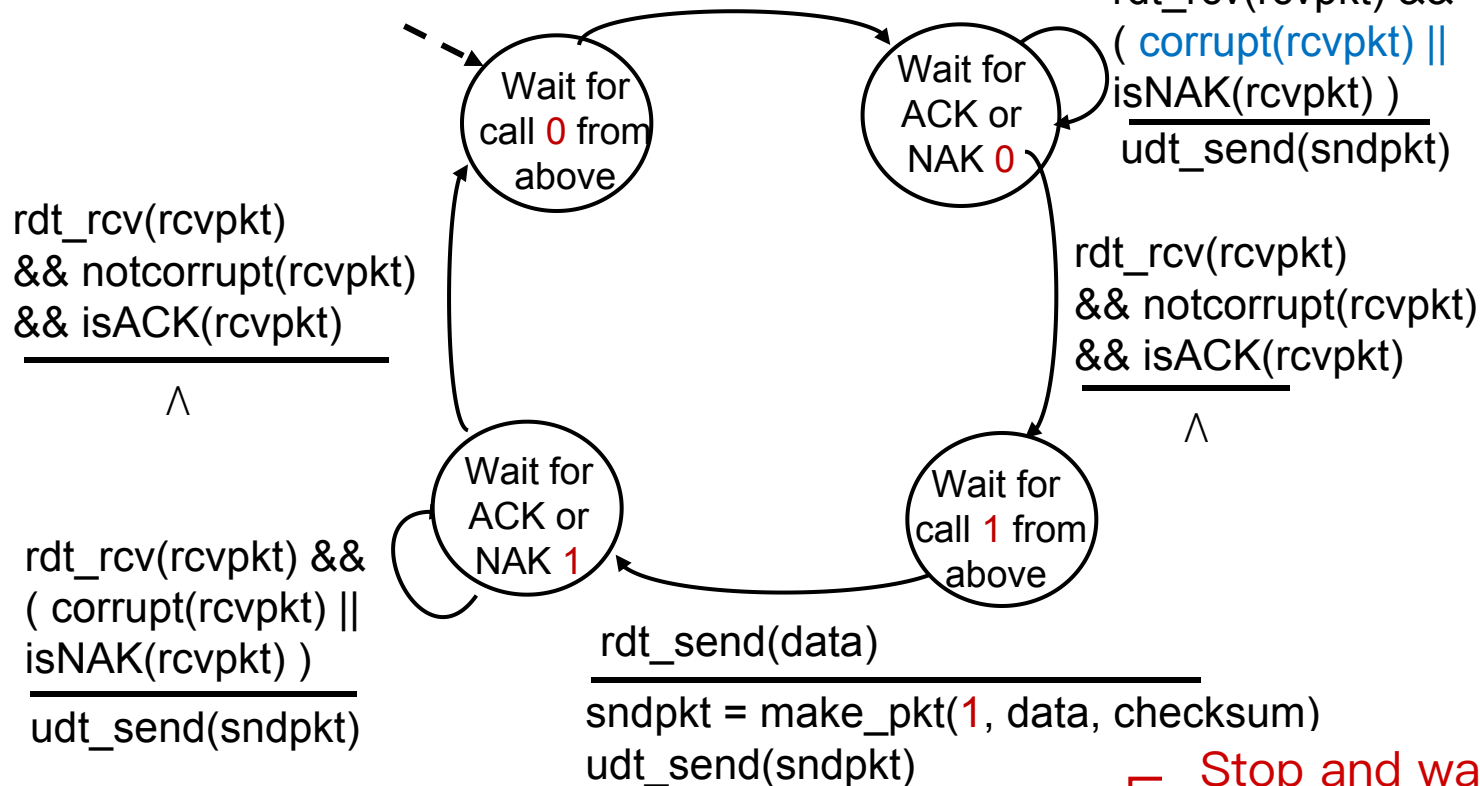
rdt2.1: sender, handles garbled ACK/NAKs

sender

rdt_send(data)

sndpkt = make_pkt(0, data, checksum)

udt_send(sndpkt)



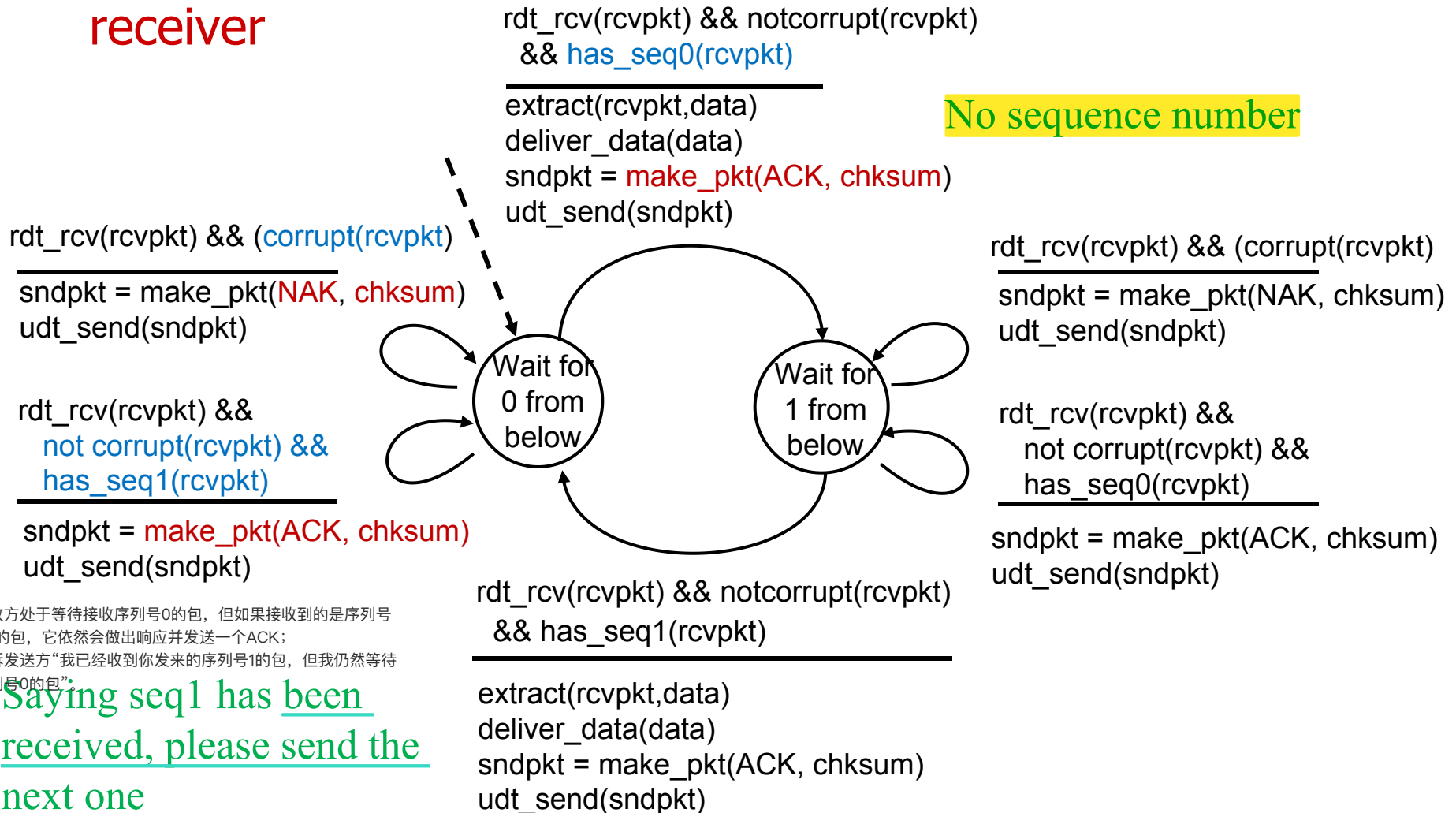
Stop and wait

Sender sends one packet, then waits for receiver response

Two sequence number would be sufficient!

rdt2.1: receiver, handles garbled ACK/NAKs

receiver



rdt2.1: discussion

sender:

- ❖ seq # added to pkt
- ❖ must check if received ACK/NAK corrupted
- ❖ twice as many states
 - state must “remember” whether “expected” pkt should have seq # of 0 or 1

receiver:

- ❖ must check if received packet is duplicate
 - state indicates whether 0 or 1 is expected pkt seq #
- ❖ note: receiver can *not* know if its last ACK/NAK received OK at sender

接收方无法知道自己刚发的 ACK/NAK 是否被发送方正确收到（反馈对反馈的问题）。因此它只能靠重复 ACK 策略来“自证立场”。

Two seq. #’s (0,1) will suffice. Why?

rdt2.2: a NAK-free protocol

当接收方发现当前包损坏或不是期望序号时：
→ 发送对“最后一次正确收到的包”的 ACK (duplicate ACK)。
(ACK 里必须显式包含被确认的序号。)

- ❖ same functionality as rdt2.1, **using ACKs only**
- ❖ instead of NAK, receiver sends ACK for last pkt received OK
 - receiver must *explicitly* include seq # of pkt being ACKed
- ❖ **duplicate ACK** at sender results in same action as NAK: *retransmit current pkt*

发送方行为：

如果收到 期望的 ACK（与当前等待的序号一致） → 正常前进、切换序号。

如果收到 重复 ACK（确认的是上一个序号）或 ACK 损坏/看不懂 → 将其视为“对当前包的否定信号”，重传当前包。

rdt2.2: sender, receiver fragments

sender

rdt_send(data)

sndpkt = make_pkt(0, data, checksum)

udt_send(sndpkt)

Wait for
call 0 from
above

sender FSM
fragment

Wait for
ACK
0

rdt_rcv(rcvpkt) &&
(corrupt(rcvpkt) ||
isACK(rcvpkt,1))
udt_send(sndpkt)

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt)
&& isACK(rcvpkt,0)

∧

receiver FSM
fragment

Wait for
0 from
below

rdt_rcv(rcvpkt) && notcorrupt(rcvpkt)
&& has_seq1(rcvpkt)

extract(rcvpkt,data)

deliver_data(data)

sndpkt = make_pkt(ACK, 1, chksum)

udt_send(sndpkt)

receiver

rdt_rcv(rcvpkt) &&
(corrupt(rcvpkt) ||
has_seq1(rcvpkt))
udt_send(sndpkt)

保持上一个ACK

只画了一半

rdt3.0: channels with errors *and* loss

New assumption: underlying channel can also **lose packets** (data, ACKs)

- checksum, seq. #, ACKs, retransmissions will be of help ... but not enough

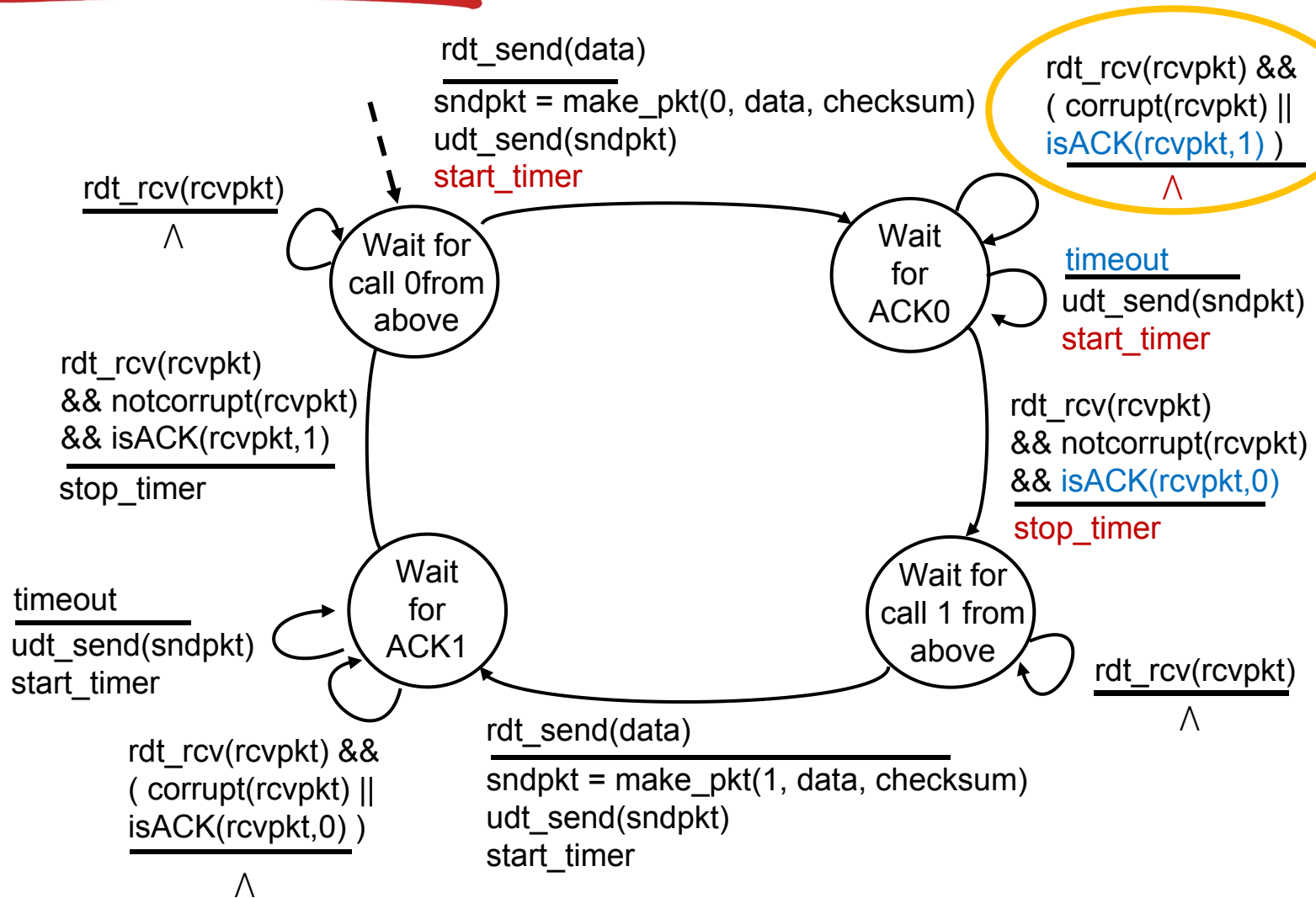
Approach: sender waits “reasonable” amount of time for ACK

- ❖ retransmits if no ACK received in this time
- ❖ if pkt (or ACK) just delayed (not lost):
 - retransmission will be duplicate, but seq. #’s already handles this
 - receiver must specify seq # of pkt being ACKed
- ❖ requires **countdown timer**
 - start timer, timer interrupt, stop timer

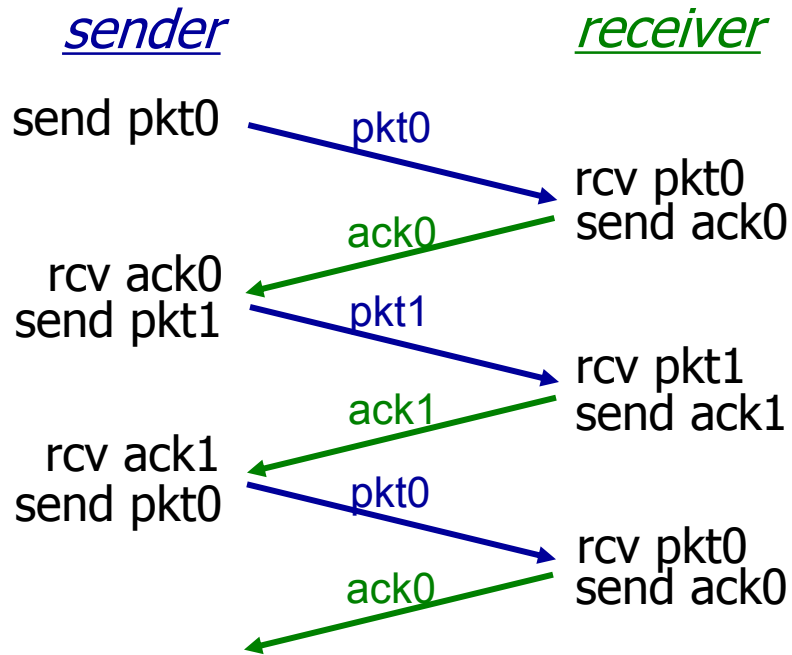
How long should the sender wait?

超时估计 (timeout estimation) ,
实际网络中 TCP 使用动态算法 (如 RTT + 安全系数) 来调整超时。

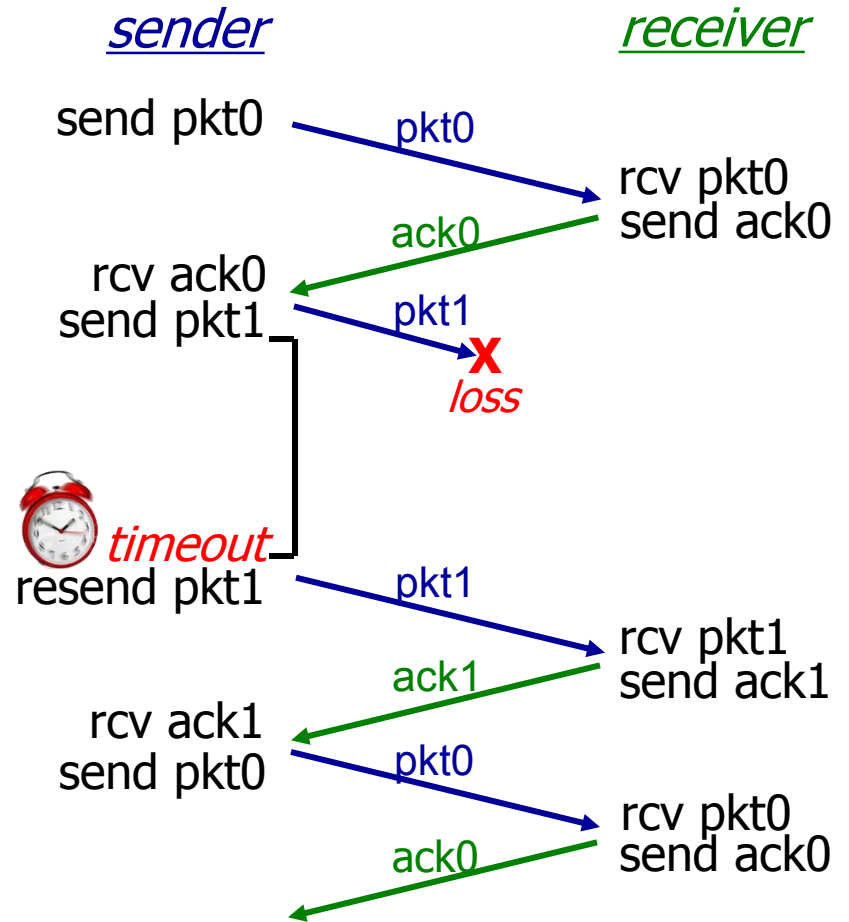
rdt3.0 sender



rdt3.0 in action

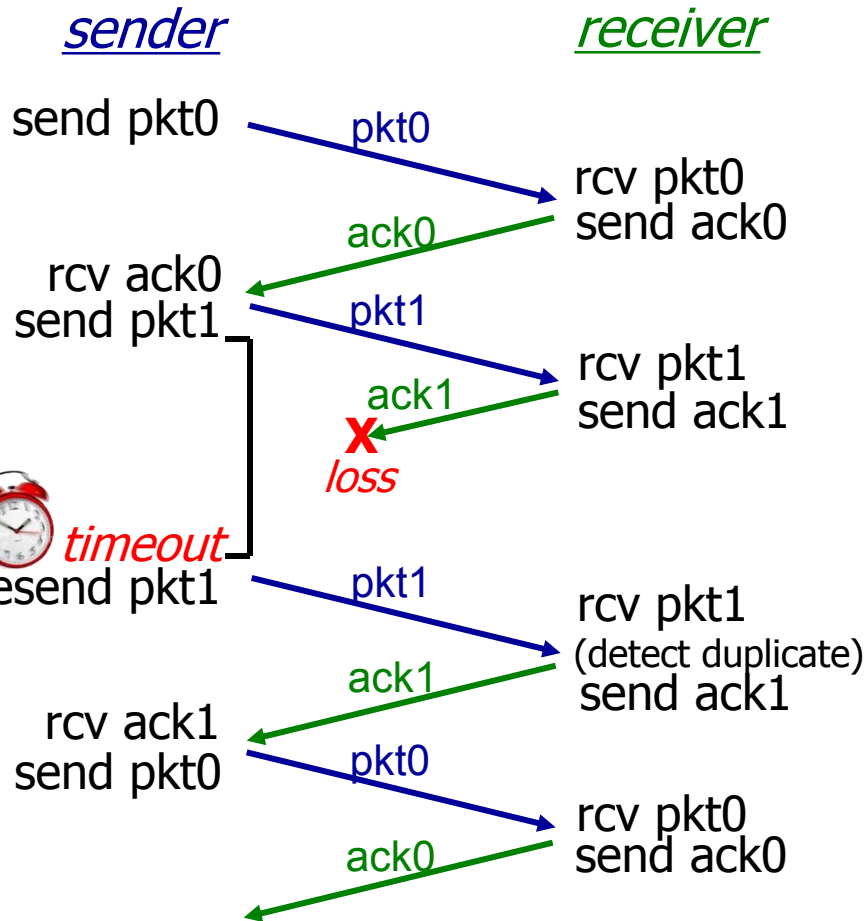


(a) no loss

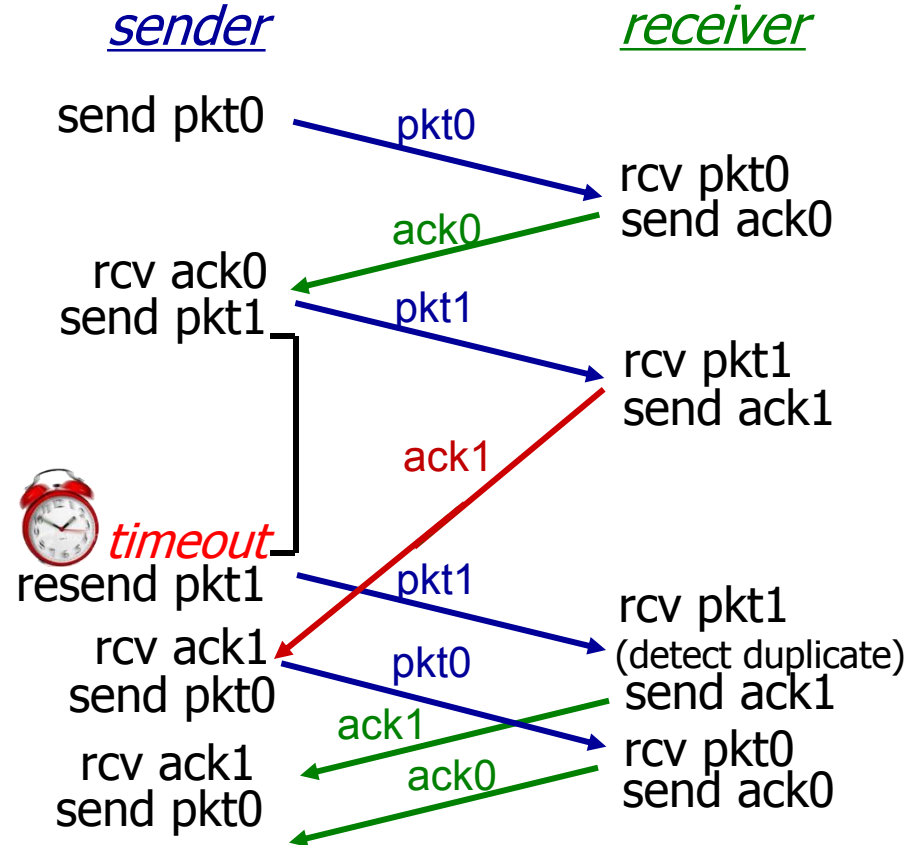


(b) packet loss

rdt3.0 in action



(c) ACK loss



(d) premature timeout/ delayed ACK

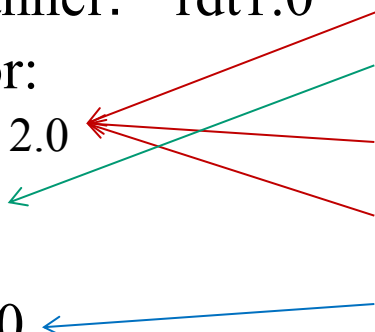
Summary

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- ❖ Channel with bit error:
 - bit error in packet: rdt 2.0
 - bit error in ACK: 2.1
 - NAK-free: 2.2
- ❖ Lossy channel: rdt 3.0

Summary of Techniques

- Checksum
- Sequence number
- ACK packets
- Retransmission
- Timeout



Next Lecture

3.1 transport-layer services

3.2 multiplexing and demultiplexing

3.3 connectionless transport: UDP

3.4 principles of reliable data transfer (continue)

3.5 connection-oriented transport: TCP

- segment structure
- reliable data transfer
- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control